

Basic 10X Genomics data analysis

Learnin goals

- Understand why filter data
- Understand why normalize data
- Understand why scale data?
- Understand why do dimensional reduction
- Know the popular dimensional reduction methods
- Understand why do clustering
- Understand why do dimensional reduction
- Understand why do cell annotation

Single cell RNAseq data analysis workflow

Preprocessing
(cellRanger)

Filtering

Normalization

Feature
selection

Scaling

PCA

Visualizaiton:
UMAP/tSNE

Gene set
analysis

Differential analysis:
Clusters/Conditions

Annotation

Multi-batch
Integration

Clustering

Cluster based annotation

Cell-cell
communication
Cellchat, Nichenet

TF
activity
SCENIC

Trajectory
PAGA
Slingshot

RNA
velocity
scvelo

Metabolic
activity
scCellFie

Cell based annotation

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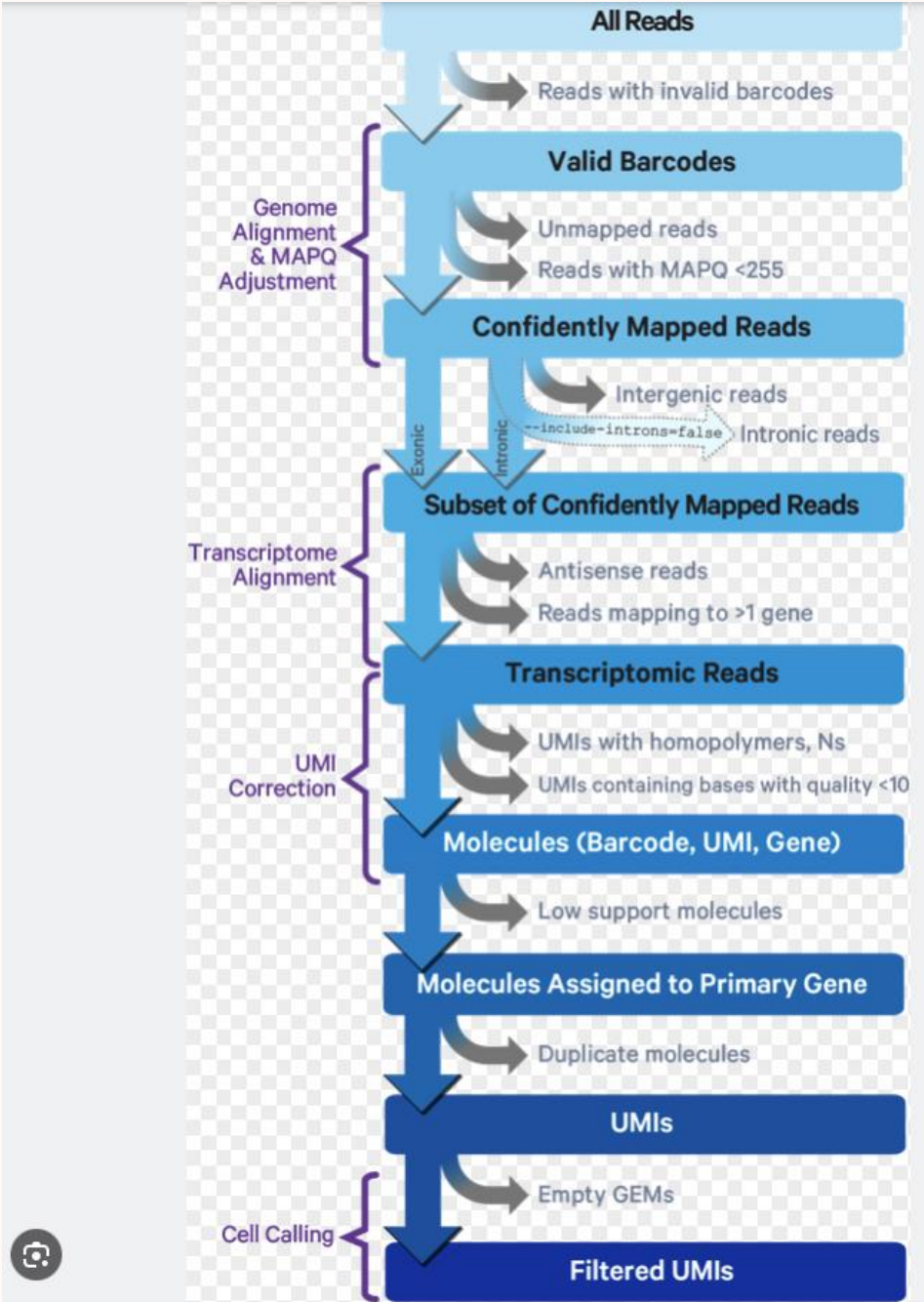
Cell based annotation

10X Genomics

Read mapping and counting

Cellranger Pipeline

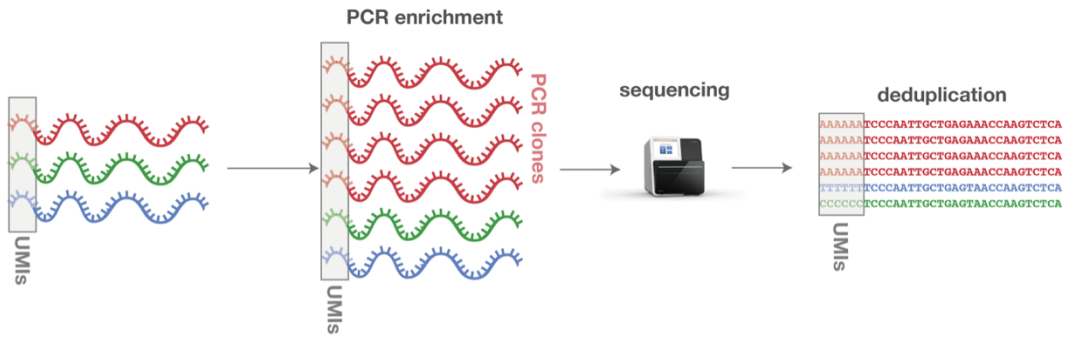
Get
gene x cell
count table



Cell Ranger's Gene Expression Algorithm - Official 10x Genomics Support

UMI: unique molecular identifiers

- Gene counts are UMI count
- Remove PCR introduced variation



<https://www.ecseq.com>

Count matrix

	Cell1	Cell2	..
Gene1	0	0	0
Gene2	0	1	0
Gene3	6	0	0
...	0	0	0

Question:

- **Single cell vs Bulk RNAseq:**
 - How they differ in mRNA quantification (apart from cell vs tissue)

Downstream analysis_popular tools

- **Seurat** (R)
- Scanpy (python)
- Scraper (R)

Data variation

Technical

- Cell quality (dying cells, broken cells...)
- Library prep efficiency (RNA capture, sequencing...)
- Batch effects
- ...

Biological

- Cell cycle
- Sex, Age
- Transcriptional bursting
- Cell size
- Cell type/state
- ...

Count matrix

	Cell1	Cell2	..
Gene1	0	0	0
Gene2	0	1	0
Gene3	6	0	0
...	0	0	0

Data variation

Technical

- Cell quality (dying cells, broken cells...)
- Library prep efficiency (cDNA capture, PCR amplification...)
- Batch effects
- ...

Biological

- Cell cycle
- Sex, Age
- Transcriptional bursting
- Cell size
- Cell type/state ✓
- ...

Count matrix

	Cell1	Cell2	..
Gene1	0	0	0
Gene2	0	1	0
Gene3	6	0	0
...	0	0	0

Remove unwanted variation
to look for biological
celltypes/states

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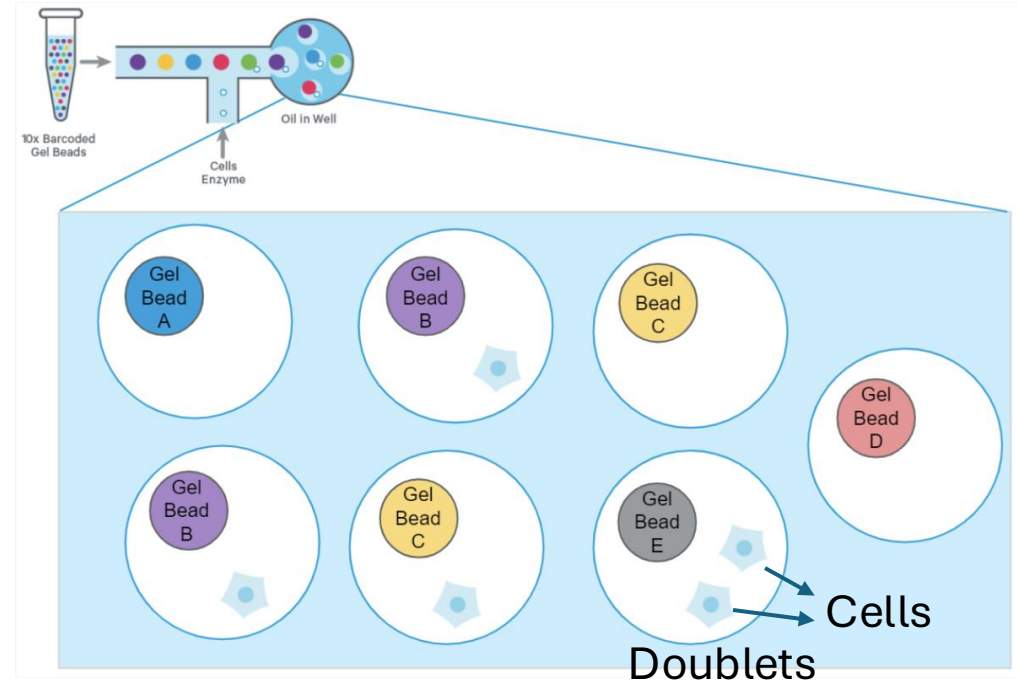
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activity
scCellFie

Cell based annotation

Filtering

- ❖ Filter out genes
 - Lowly captured genes (e.g. ≤ 3 cells)
- ❖ Filter out RNAs
 - Cell-free, 'ambient' RNA within cell suspension
- ❖ Only keep viable cells
 - Empty droplets (few genes/UMI)
 - Low viable cells (few genes/UMI)
 - **Doublets (too many genes/UMI)**
 - Broken cells (leaking)
 - High mitochondrial gene percentage & few genes/UMI
 - Cytosolic transcripts easier to leak out than mitochondria (big)

Doublet: a 10X droplet capturing 2 cells



<https://kb.10xgenomics.com>

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Normalization

Sampling effect in

RNA capture by beads
RNA $\longrightarrow$ cDNA
Sequencing of cDNA

Cell cycle
Cell size
Transcriptional burst
Batch effect



Total cell UMI count
differ among cells

Normalization

Sampling effect in

RNA capture by beads
RNA $\rightarrow$ cDNA
Sequencing of cDNA

Cell cycle

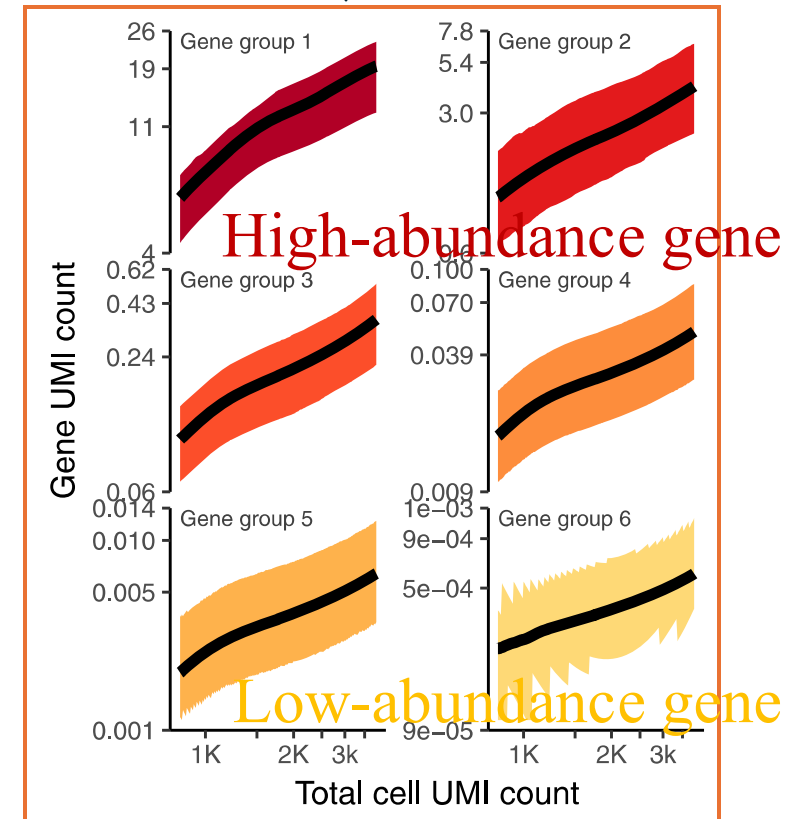
Cell size

Transcriptional burst

Batch effect



Total cell UMI count
differ among cells



➤ Gene abundance depends on total cell UMI count
(desired biological differences?)



Normalization

Sampling effect in

RNA capture by beads
RNA $\rightarrow$ cDNA
Sequencing of cDNA

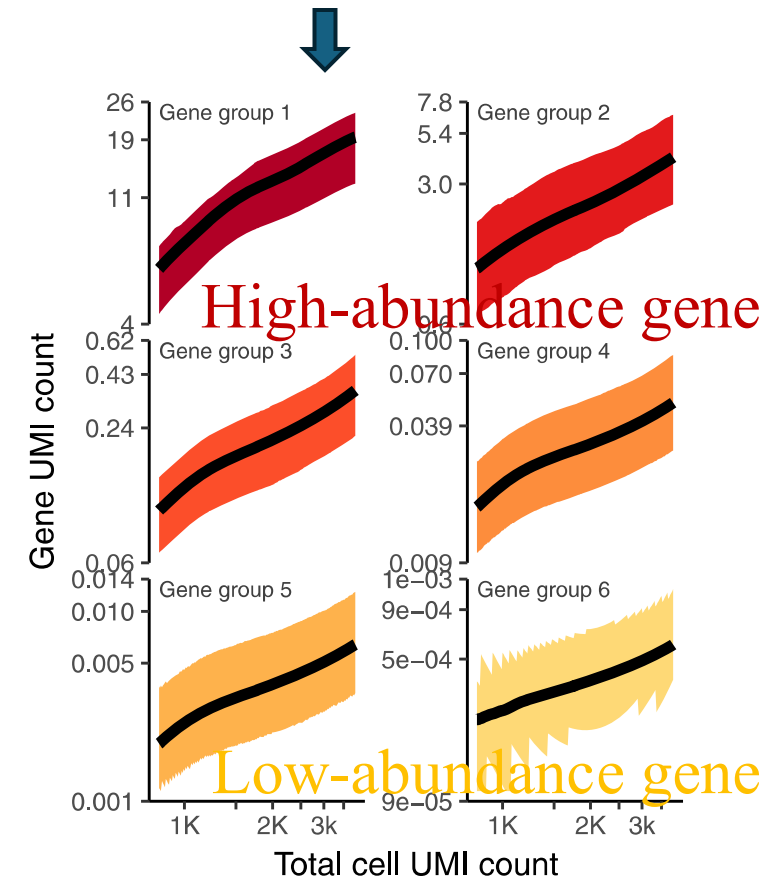
Cell cycle
Cell size
Transcriptional burst
Batch effect

➤ Gene abundance depends on total cell UMI count
(desired biological differences?)

➤ Normalization

- (try to) Remove total UMI count differences among cells
- Make gene expression comparable between cells

Total cell UMI count
differ among cells



Normalization methods

➤ Seurat: LogNormalize (Scaling)

* *Assume all cells with the same molecular composition*

RNA:count slot: raw count

Seurat::NormalizeData()

Each cell's counts are:

1. Divided by the total counts for that cell (to no
2. Multiplied by a scale factor (default = 10,000)
3. Log-transformed (using natural log).

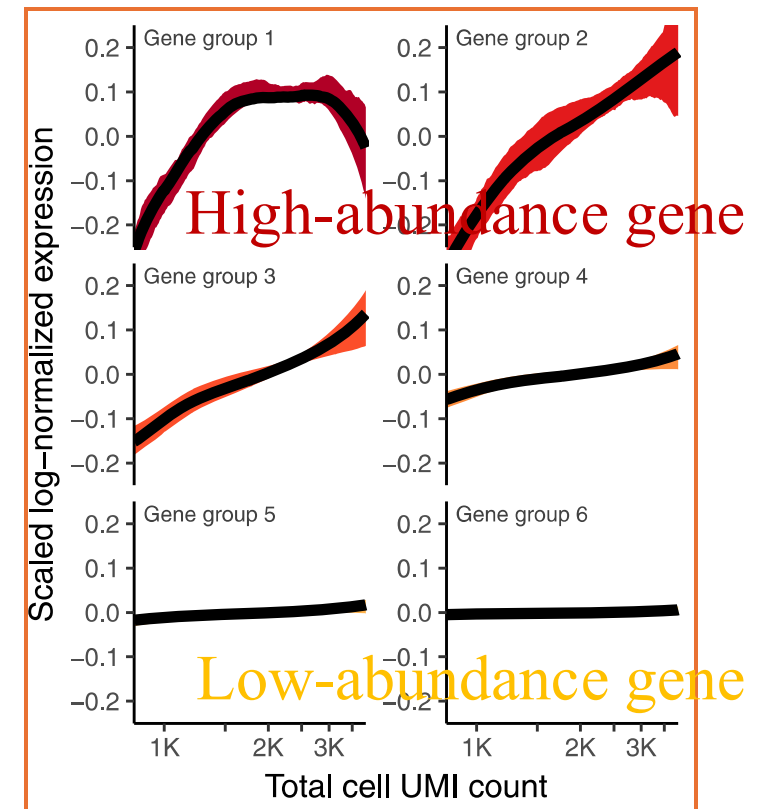
$$x'_{ij} = \log \left(\frac{x_{ij}}{\text{total}_j} \times 10,000 + 1 \right)$$

- x_{ij} : raw count of gene i in cell j
- total_j : total counts in cell j

RNA:data slot:
* Normalized count
* Near normal distribution
* All cells the same total UMIs

$$x_i = \frac{\text{The read count of gene } X \text{ in cell } i}{\text{Total counts of cell } i} \times 10^4 \text{ (Eq.1)}$$

$$f(x_i) = \ln(x_i + 1) \text{ (Eq.2)} \quad \text{More or less}$$



Normalization methods

- Seurat: LogNormalize (Scaling)

- * Assume all cells with the same molecular composition

- High-count filtering CPM

- BASiCs (Spike-in based)

- Scrان (Lun et al. 2016) (Pooling-based)

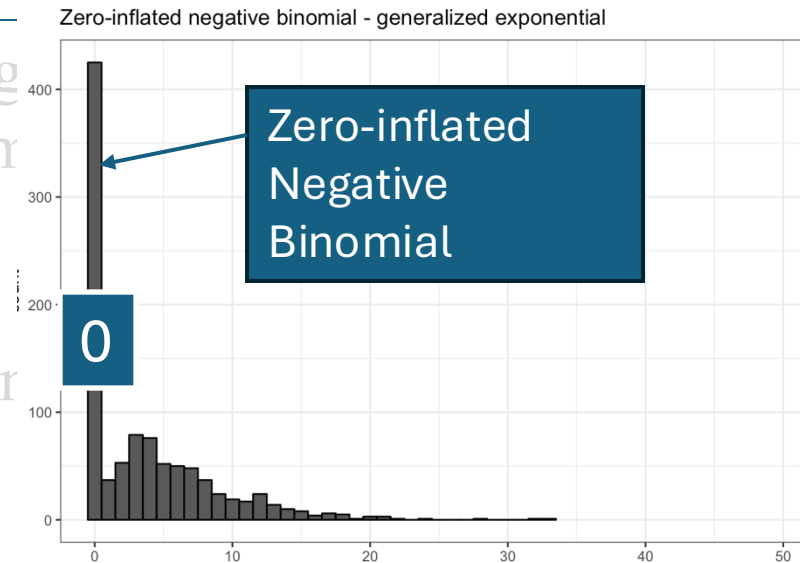
- **Downsampling**

Normalization methods

- Seurat: LogNormalize (Scaling)
 - * Assume all cells with the same total number of reads
- High-count filtering CPM
- BASiCs (Spike-in based)
- Scraper (Lun et al. 2016) (Pooling)
- Downsampling

➤ ZINB-Wave (**overfitting**)

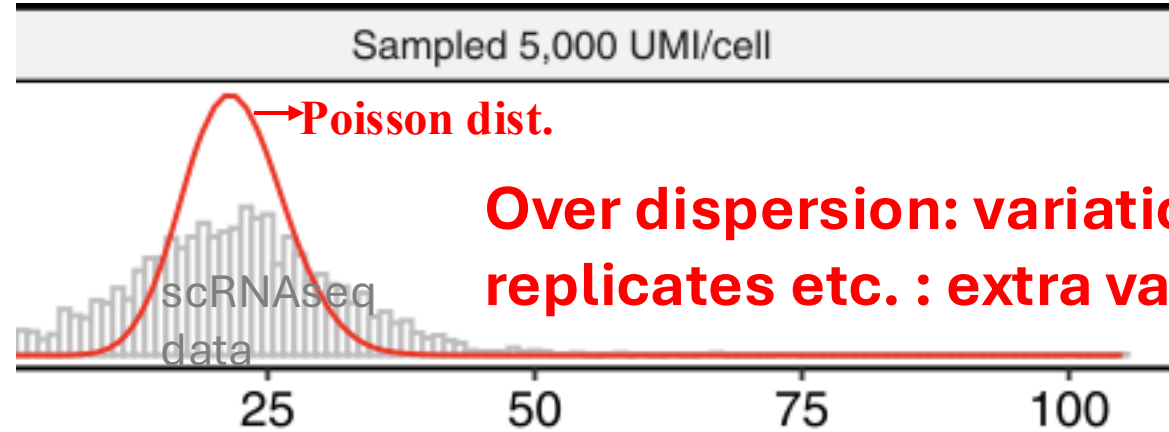
➤ **Regularized Negative Binomial Regression/Seurat: SCTransform v2**



Model parameters may differ between datasets: normalize separately

Single cell RNAseq data modeling:

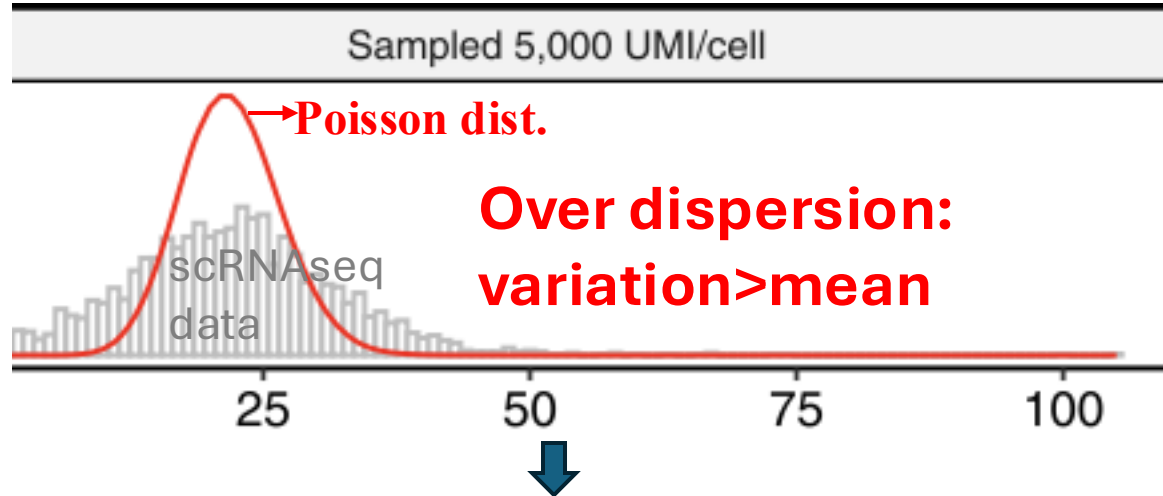
❖ Count: poisson distribution (variation=mean)



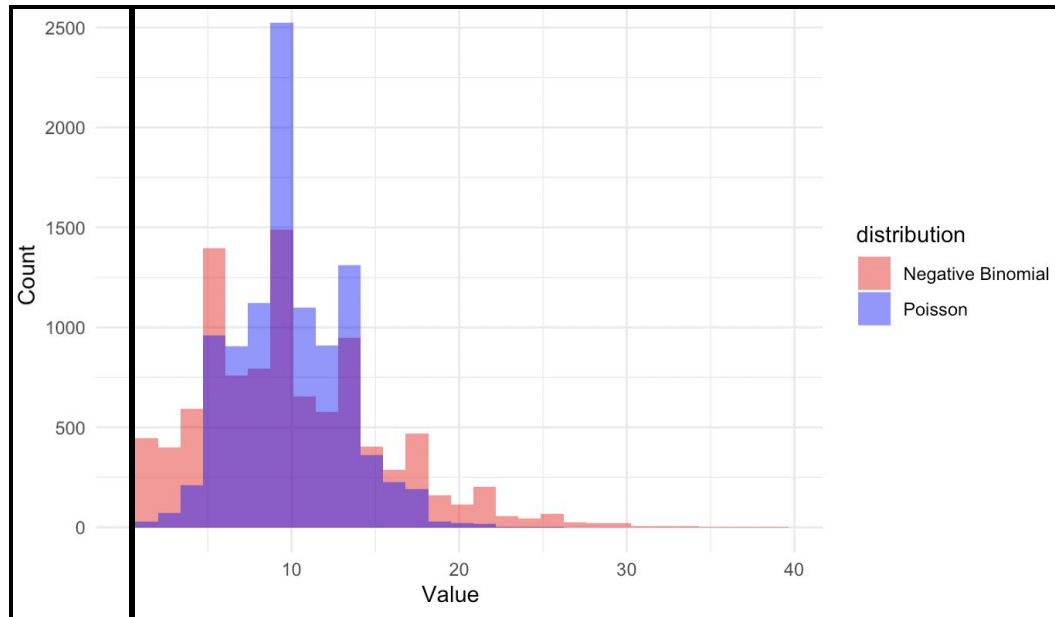
Over dispersion: variation > mean
replicates etc. : extra variation to count data

Single cell RNAseq data modeling:

❖ Poisson distribution

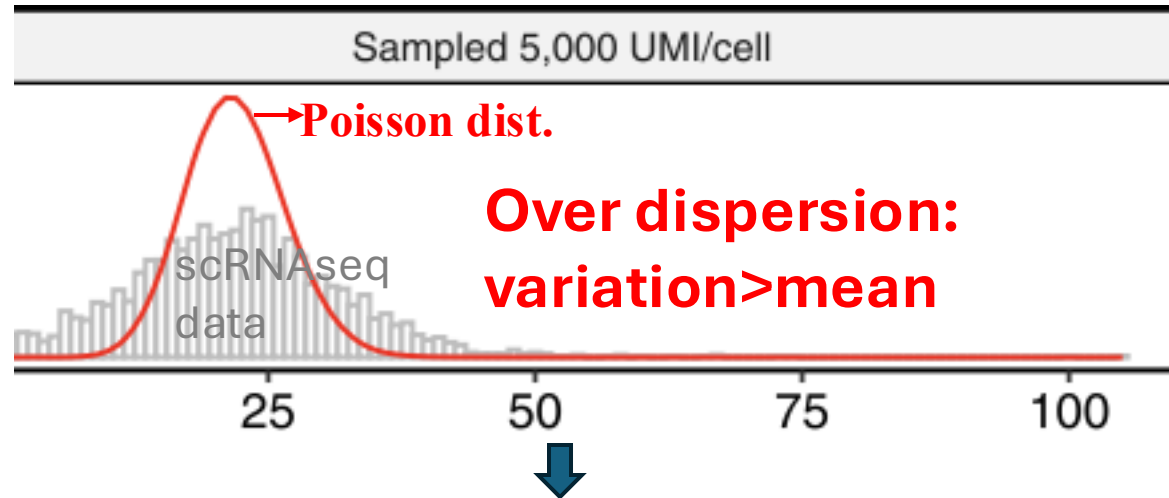


❖ Negative Binomial

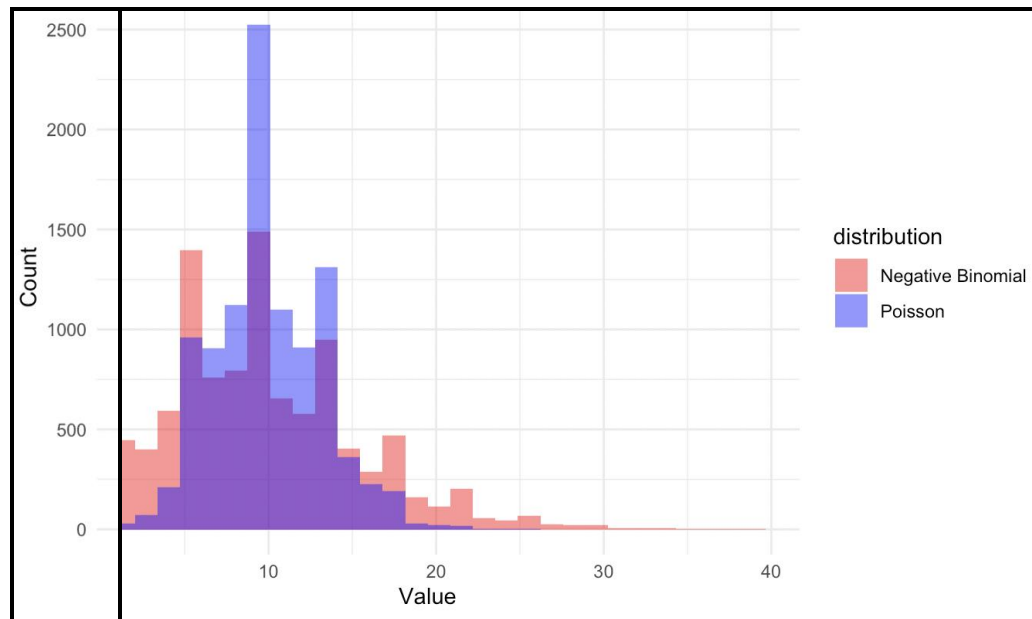


Single cell RNAseq data modeling:

❖ Poisson distribution



❖ Negative Binomial



NB GLM:

$$\log(\mathbb{E}(x_i)) = \beta_0 + \beta_1 \log_{10} m_i$$

$$y_i = \text{NB}(\mathbb{E}(x_i), \theta)$$

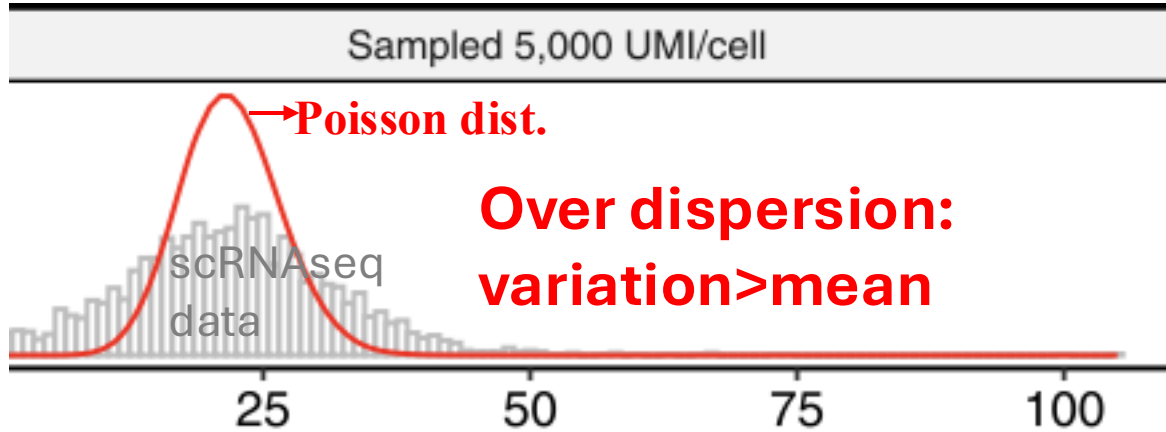
↓
**Dispersion
parameter**

Total cell UMI

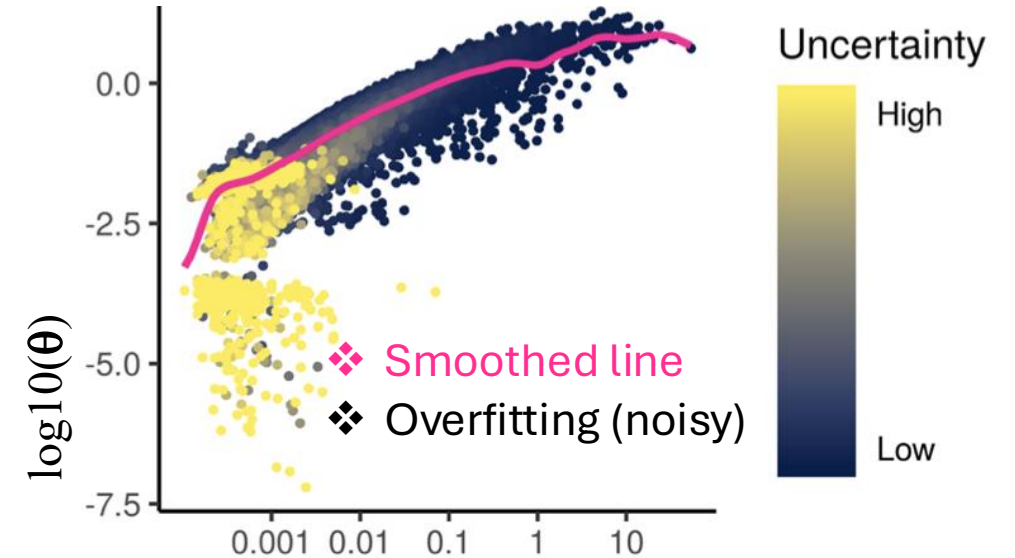
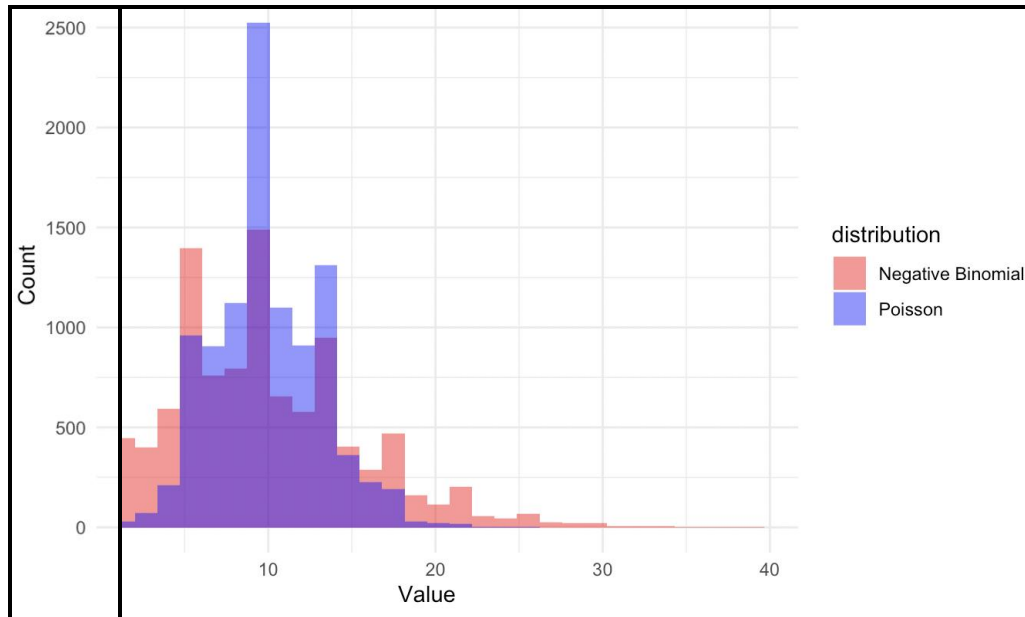
Remove total cell
UMI dependence

Single cell RNAseq data modeling:

❖ Poisson distribution (variation=mean)



❖ Negative Binomial



➡

NB GLM:

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$y_i = \text{NB}(\mathbb{E}(x_i), \theta)$

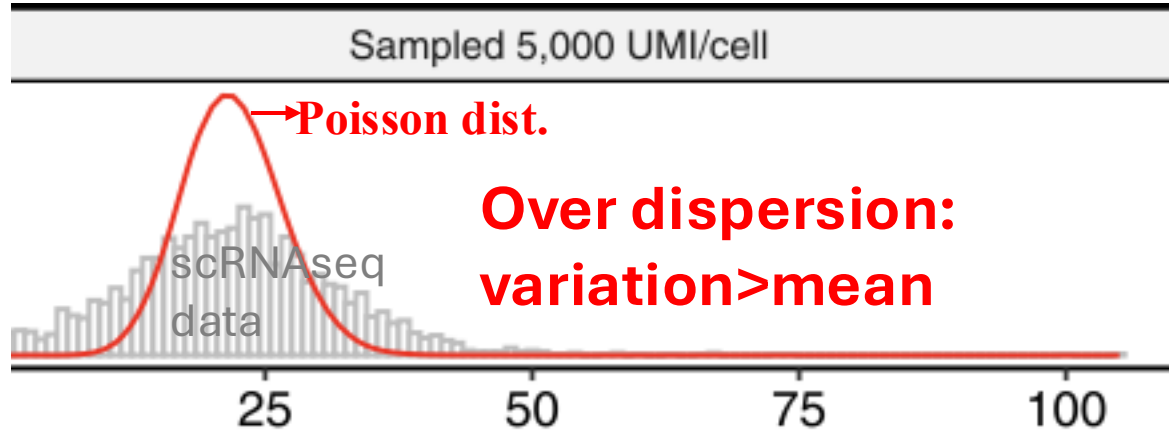
Dispersion parameter

Total cell UMI

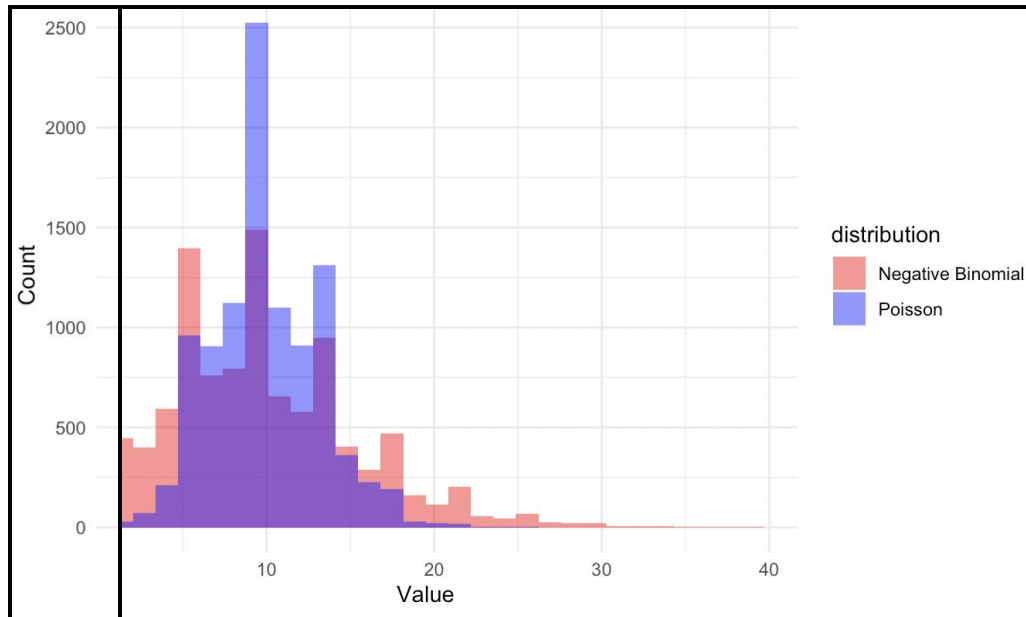
Remove total cell UMI dependence

Single cell RNAseq data modeling:

❖ Poisson distribution (variation=mean)

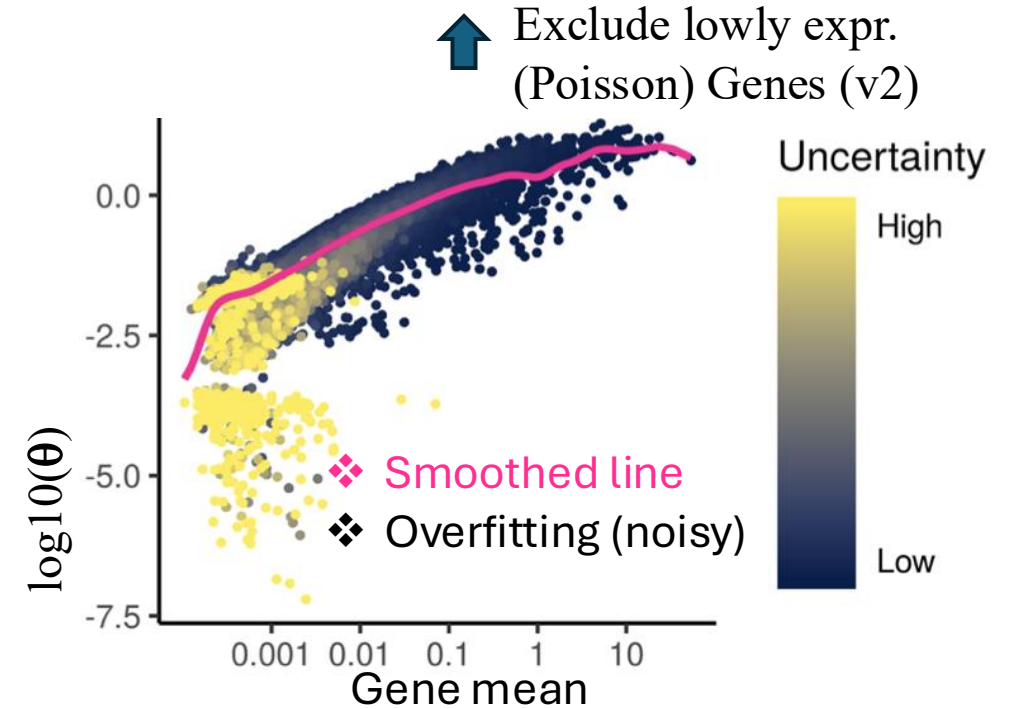


❖ Negative Binomial



Model parameter Regularization

- * Constrain towards smoothed line
- * Increase estimate robustness



NB GLM:

$$\log(\mathbb{E}(x_i)) = \beta_0 + \beta_1 \log_{10} m_i$$

$$y_i = \text{NB}(\mathbb{E}(x_i), \theta)$$

↓
Dispersion
parameter

Total cell UMI

Remove total cell
UMI dependence

NB GLM:

$$\log(\mathbb{E}(x_i)) = \beta_0 + \beta_1 \log_{10} m_i + \text{Covariates}$$

$$y_i = \text{NB}(\mathbb{E}(x_i), \theta)$$

Dispersion
parameter

Remove total cell
UMI dependence

Total cell UMI



y_{ij} observed count for gene i in cell j

μ_{ij} expected mean for gene i , cell j

β_{0i} intercept (baseline mean)

$$\hat{\mu}_{ij} = \exp(\beta_0 + \beta_1 \log(\text{UMI}_j))$$



Pearson residual:

$$r_{ij} = \frac{y_{ij} - \hat{\mu}_{ij}}{\sqrt{\hat{\mu}_{ij} + \hat{\mu}_{ij}^2 / \theta_i}}$$

NB SD



SCT:data slot:

- * log transformed
- * near normal
- * total UMIs (+covariates) effect removed



$$y_{ij}^{\text{corrected}} = r_{ij} * \text{NB SD} + \exp(\beta_0)$$

SCT:count slot:

- * Neg. binomial distribution
- * overdispersion (variance > mean)
- * Variance increase with mean
- * total UMIs (+covariates) effect removed

SCT:scale.data slot:

- * No overdispersion
- * Nearly normal
- * mean = 0, variance stabilized
- * total UMIs (+covariates) effect removed
- * Ready for PCA

Dimensional reduction

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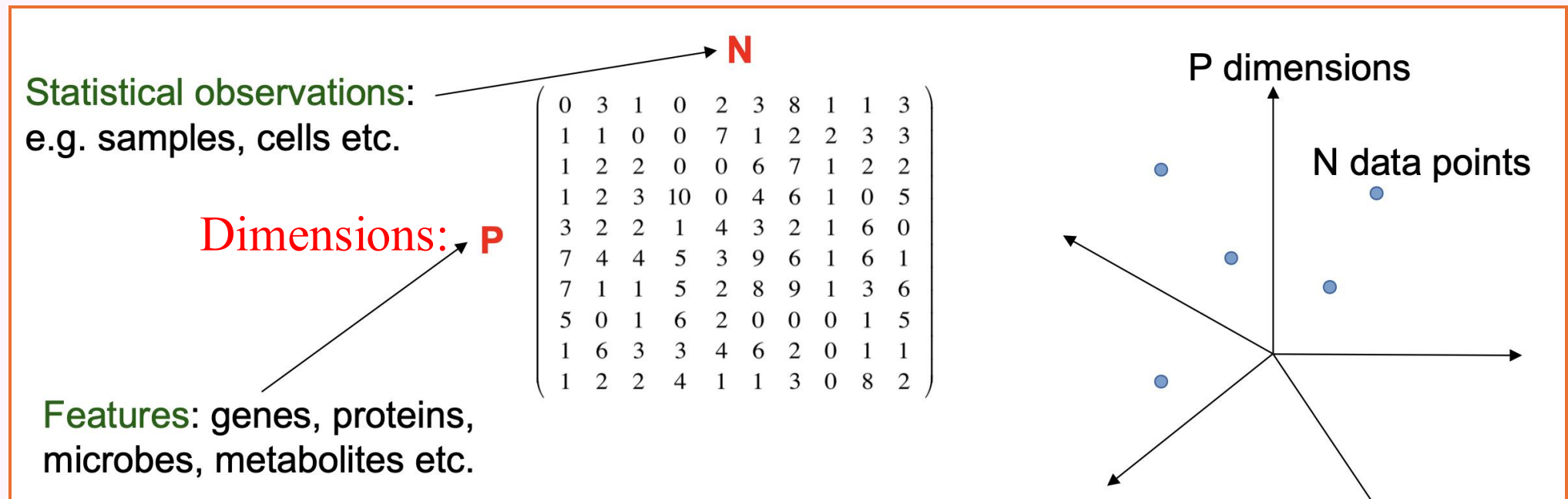
Trajectory
PAGA
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High dimensional curse

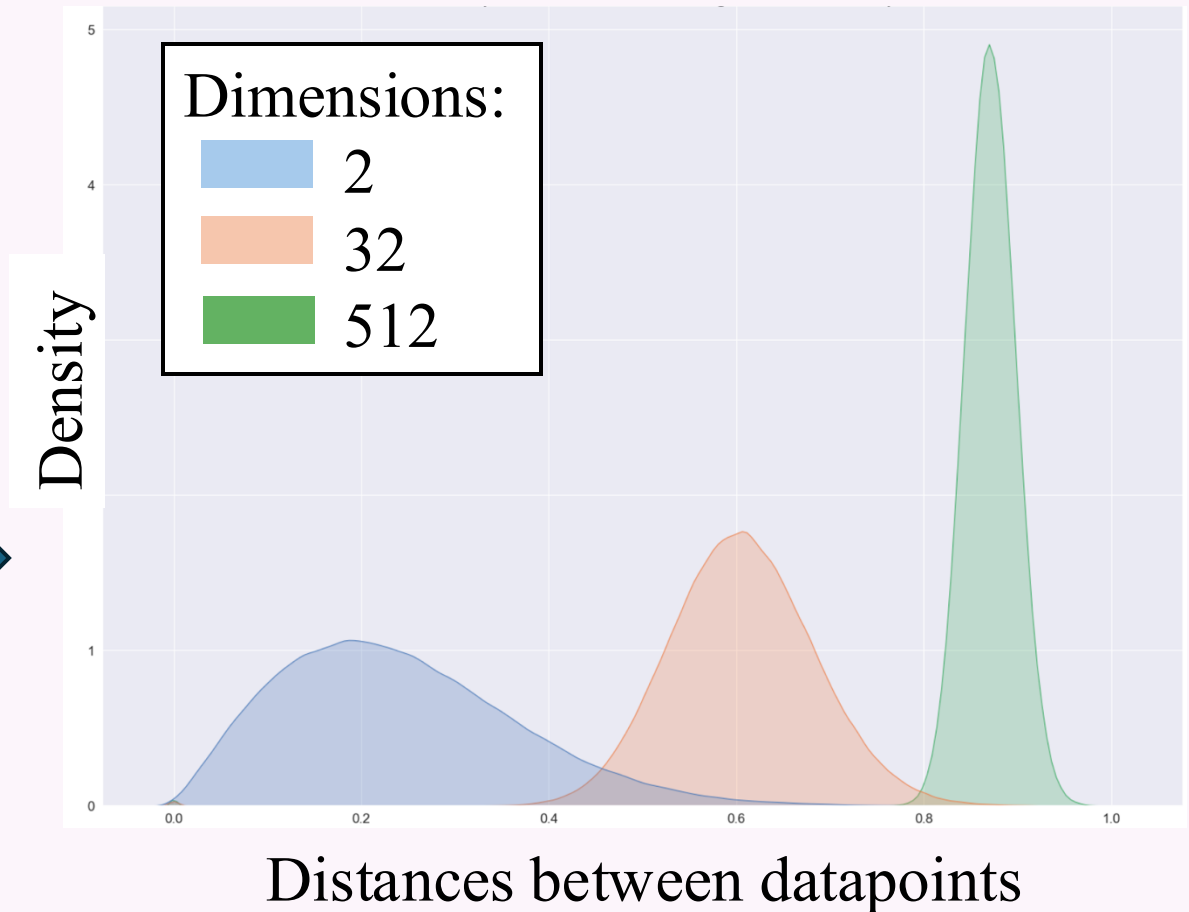
Single cell data: high dimensional data (**P genes = P dimesions, thousands of genes**)



High dimensional curse

Curses:

- Data sparsity (missing values, noisy)
- Multicollinearity (correlation between genes)
- Multiple testing
- Overfitting
- Data points very far from each other
(**hard to form clusters/celltypes!**
Visualization)



High dimensional curse

❖ Solutions:

➤ Select (e.g. 2k) highly variable genes/HVGs (less genes/dimensions)

➤ **Dimensional reduction**

(Compress 2k HVGs into **fewer** dimensions/`MegaGenes`)

* (Scaling)

- PCA
- tSNE
- UMAP

Select highly variable genes

Assumptions:

- Features (i.e. genes) with low variance: **noises**
- Features with high variance: biological factors

➤ Seurat:sctransform()

- ❖ Based on the residual variance

➤ Seurat:FindVariableFeatures()

❖ Vst:

- Model the mean and variance of the normalized data,
- Select those features deviating the most from the predicted variance

❖ Dispersion:

- Normalized by mean

Scaling

➤ Why do scaling?

- ❖ PCA/tSNE/UMAP: multivariate analysis
- ❖ Require features to be on similar scales

➤ Seurat SCT:scale.data

- ❖ Pearson residuals (variance stabilized residuals)

➤ Seurat RNA:scale.data

- ❖ With vars.to.regress
 - Standardized raw residuals
- ❖ Without vars.to regress
 - Standardized normalized count (mean=0, variance=1)

RNA:count slot: raw count



Seurat::NormalizeData()

Each cell's counts are:

1. Divided by the total counts for that cell (to no)
2. Multiplied by a scale factor (default = 10,000)
3. Log-transformed (using natural log).

$$x'_{ij} = \log \left(\frac{x_{ij}}{\text{total}_j} \times 10,000 + 1 \right)$$

- x_{ij} : raw count of gene i in cell j
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RNA:data slot: normalized count

- * Near normal distribution
- * All cells the same total UMIs



RNA:scale.data slot:

- * mean = 0, variance=1
- * total UMIs (+covariates) effect removed



With *vars.to.regress*

$$z_{ij} = \frac{x'_{ij} - \bar{x}_i}{s_i}$$

$= \mathbf{y} - \hat{\mathbf{y}}$

- Centered to mean=0
- Scale to variance=1

Without *vars.to.regress*

Expression _{i} = $\beta_0 + \beta_1 \times \text{Var}_1 + \beta_2 \times \text{Var}_2 + \dots + \epsilon_i$

Covariates: e.g. mito

Get raw residual:

Standardize the residuals:

- * center: mean=0
- * scale: variance=1

Seurat::ScaleData()



NB GLM:

$$\log(\mathbb{E}(x_i)) = \beta_0 + \beta_1 \log_{10} m_i + \text{Covariates}$$

$$y_i = \text{NB}(\mathbb{E}(x_i), \theta)$$

Dispersion
parameter

Remove total cell
UMI dependence

Total cell UMI



y_{ij} observed count for gene i in cell j

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NB SD



SCT:scale.data slot:

- * No overdispersion
- * Variance not increase with mean
 - mean = 0, variance stabilized
- * total UMIs (+covariates) effect removed
- * Ready for PCA



SCT:data slot:

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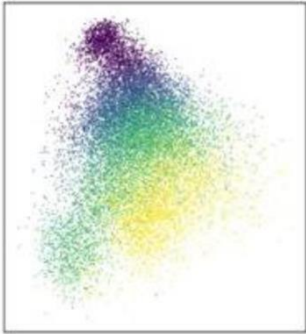
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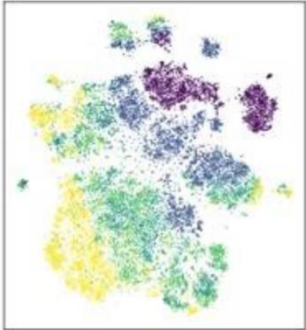
Dimensional reduction

Embryoid Body
(n=16,825)

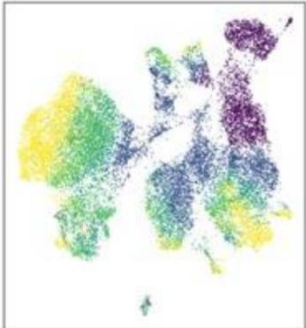
PCA



tSNE



UMAP



➤ PCA:

- Linear transf.(accurately represent distances between cells)
- Cannot efficiently pack differences in high dim. into the first 2 PCs (principal components)

Quantification

➤ tSNE

- Preserve local structure
- Perplexity, low: finer structure
- Slow

➤ UMAP:

- Preserve both local & global structure
- `n_neighbors`, `min_dist`, low: finer structure
- Fast

➤ tSNE&UMAP

- Non-linear (discard inf. distort distance)
- NOT accurately represent distances between cells

Only for visualization

What if there are multiple samples from
different batches?

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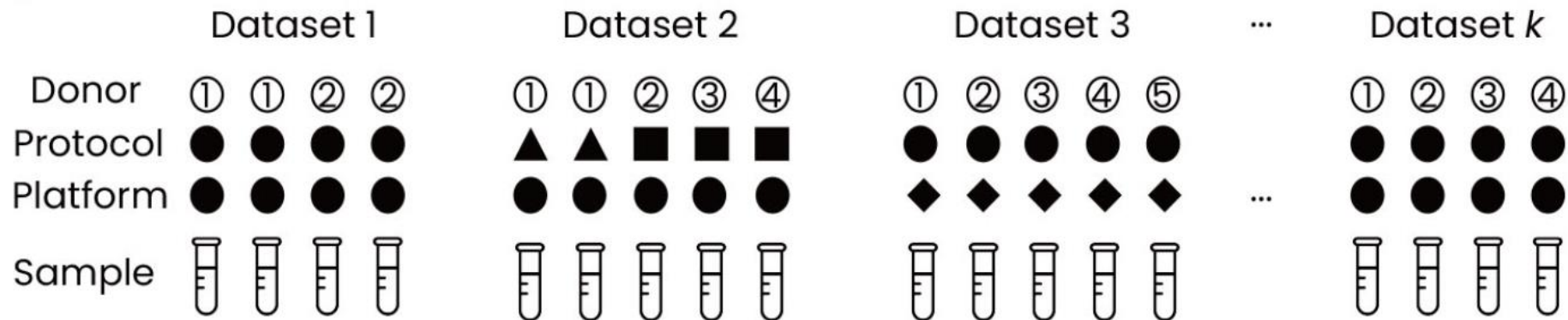
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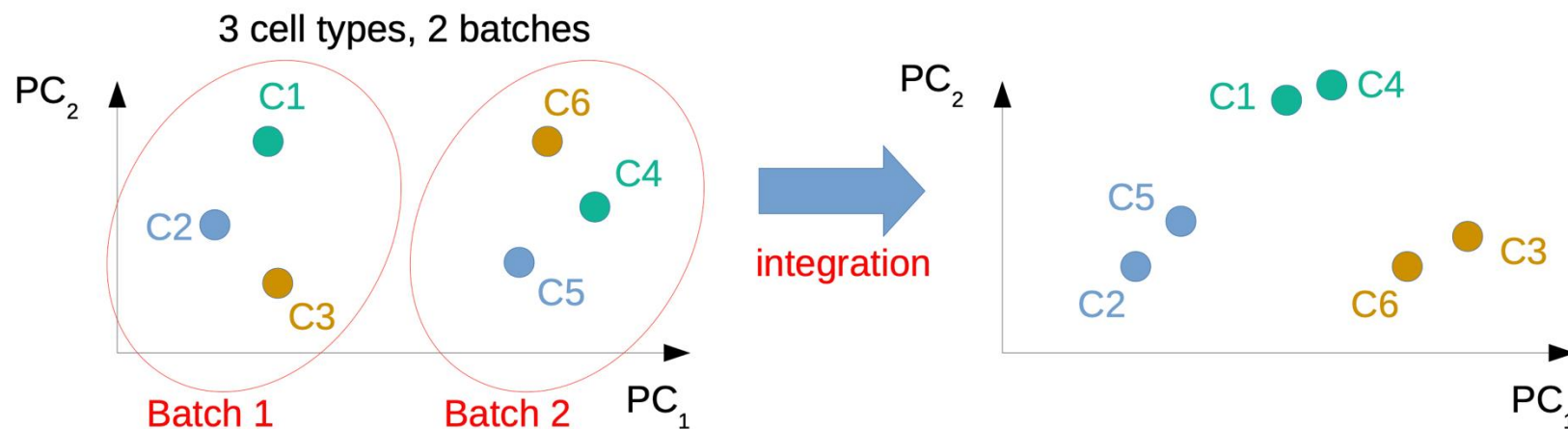
Integration: remove batch effect

Batches:

- Sample characteristics (donors, tissues, species, disease conditions) ...
- Experimental protocols, laboratories, sample acquisition and handling, sample composition, reagents or media and/or sampling time ...
- Sequencing platforms/lanes/flow cells ...



Integration: remove batch effect



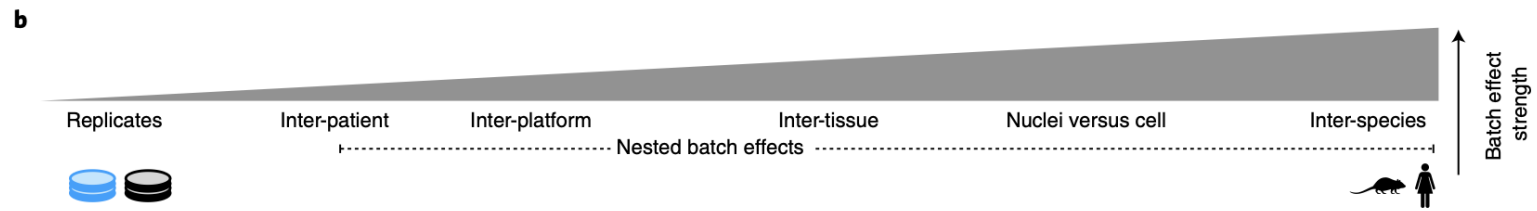
a

	Considerations	scANVI	Scanorama embed	scVI	FastMNN embed	scGen	Harmony	FastMNN gene	Seurat v3 RPCA	BBKNN	Scanorama gene	ComBat	MNN	Seurat v3 CCA	trVAE	Conos	DESC	LIGER	SAUCIE embed	SAUCIE gene
Input	Programming language																			
	Method runs without additional information	✗				✗														
Scib results	Consistent top performer	✓	✓	✓		✓														
	Top method on small/simple tasks		✓		✓	✓	✓													
	Top method on large/complex tasks	✓	✓	✓		✓														
	Top method on ATAC data	—		—			✓											✓		
Task details	Integrates strong batch effects	✓	—	—		✓			—	—				—						
	Top method for recovery cell states or modules	✓	✓								✓	✓	✓							
	Confounding of bio and batch variance	✓	—			✓														
	Top method for trajectories	—	✓	—	✓	✓														
	Method deals with varying compositions											✗								
Speed	Fast method for quick results									✓		✓								
	Scales well to large datasets on CPU	✓	—	✓						✓	—								✓	✓
	Method has GPU support	✓		✓		✓									✓		✓		✓	✓
	Scales well to feature spaces beyond genes														✓	✓				
Output	Method shows corrected expression					✓		✓	✓		✓	✓	✓	✓						✓
	Method gives relative cell embeddings									✗						✗				

Fulfills the criterion
 Python
 R

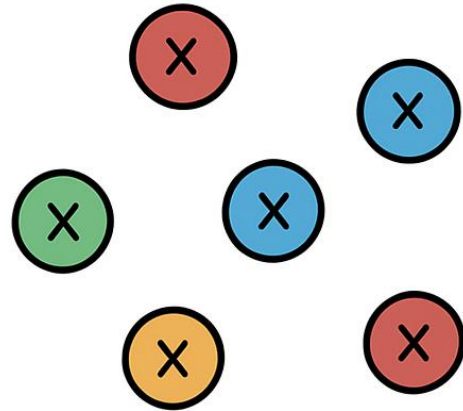
Partial fulfillment of criterion

Does not fulfill criterion



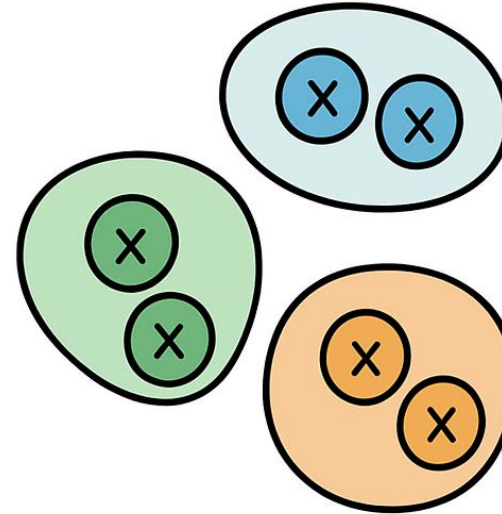
Graph-based Clustering

Analyze ind. cells?



- Noise, variability
- Biological context lost

Group cells to clusters!



- Reduce noise, robust insights
- Biologically meaningful patterns

Cell clusters/groups $\approx$ Celltypes/cellular states

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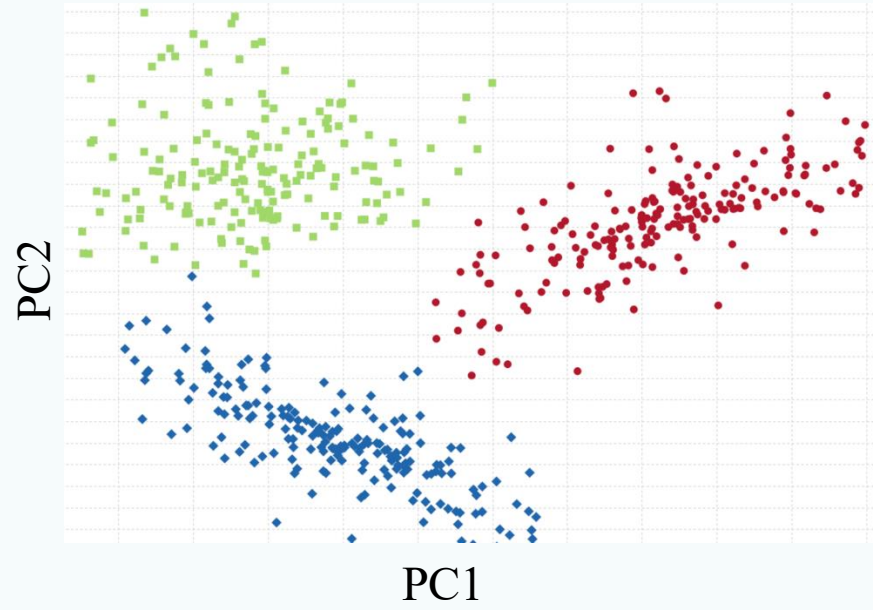
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velocity
scvelo

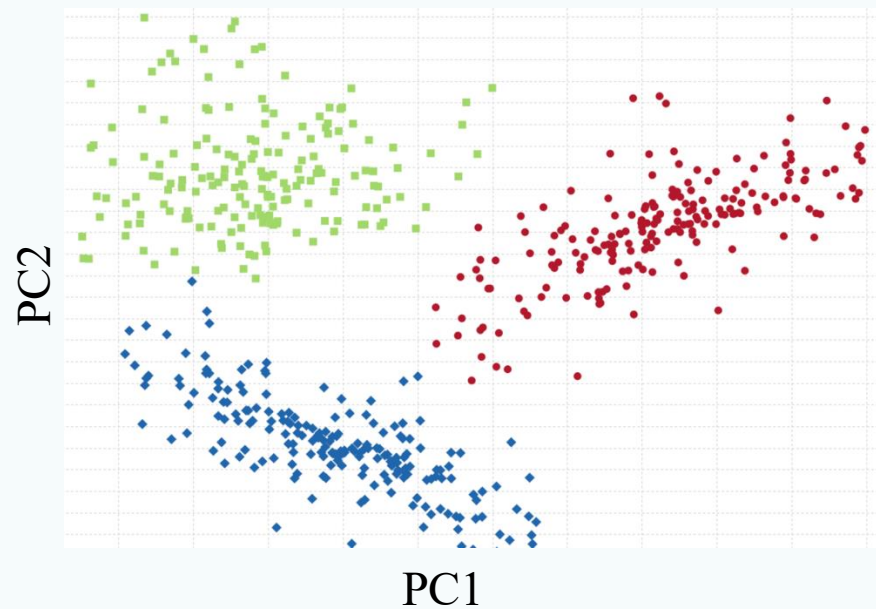
Metabolic
activity
scCellFie

Cell based annotation

PCA space: on top n PCs



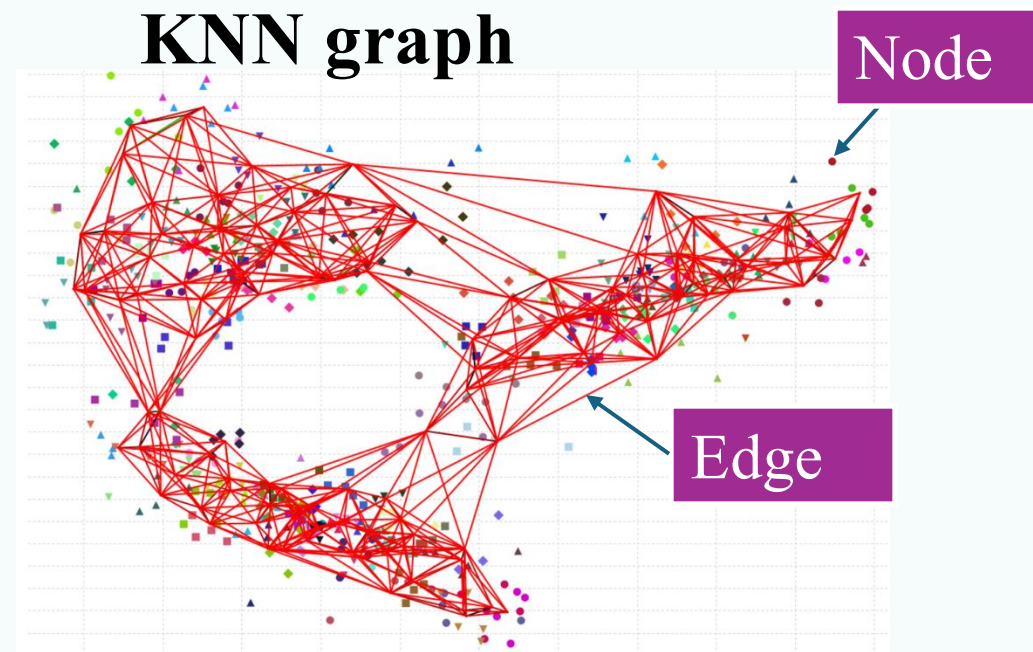
PCA space: on top n PCs



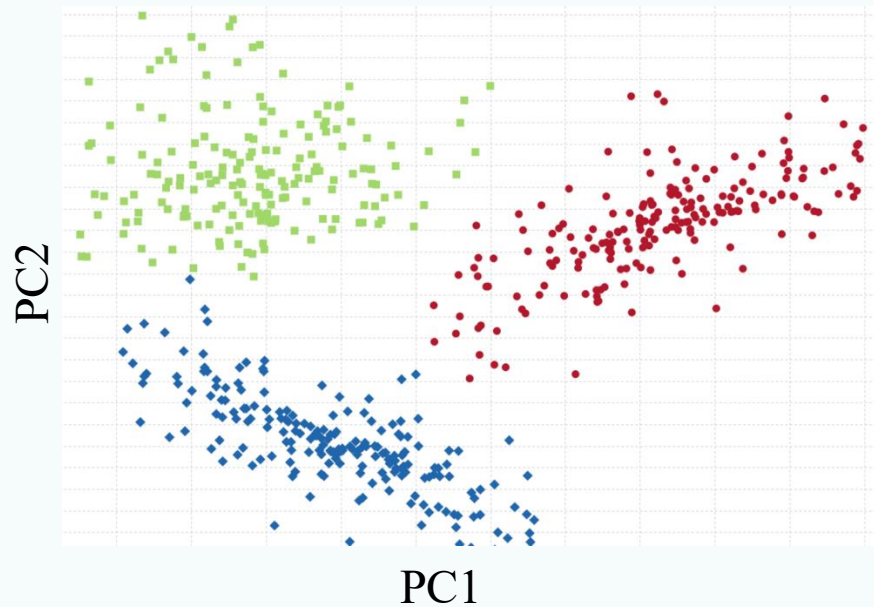
Connect K
nearest
neighbors



KNN graph



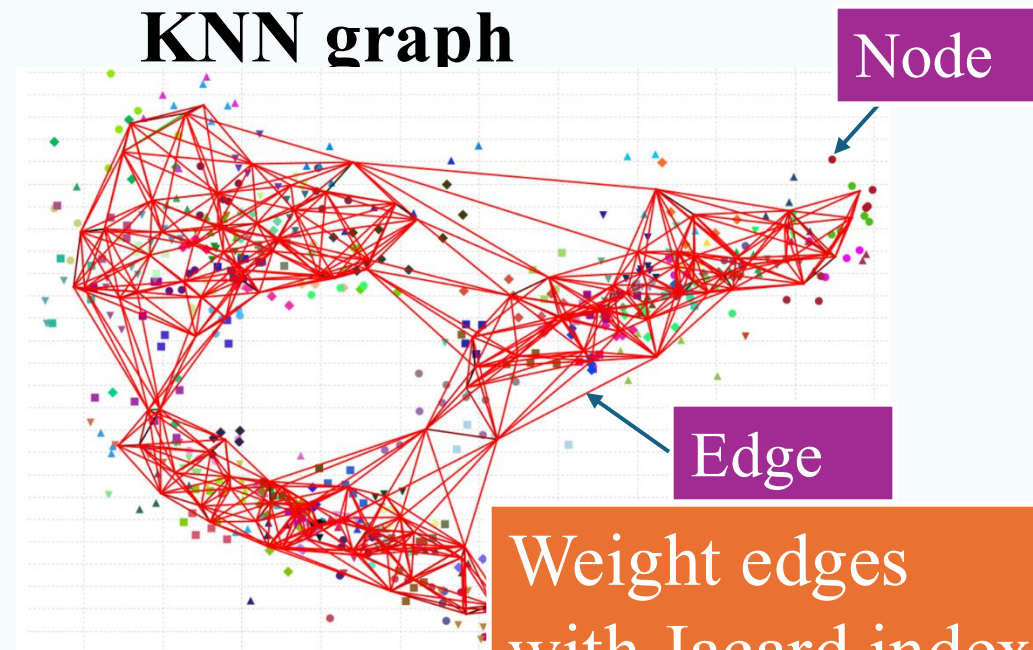
PCA space: on top n PCs



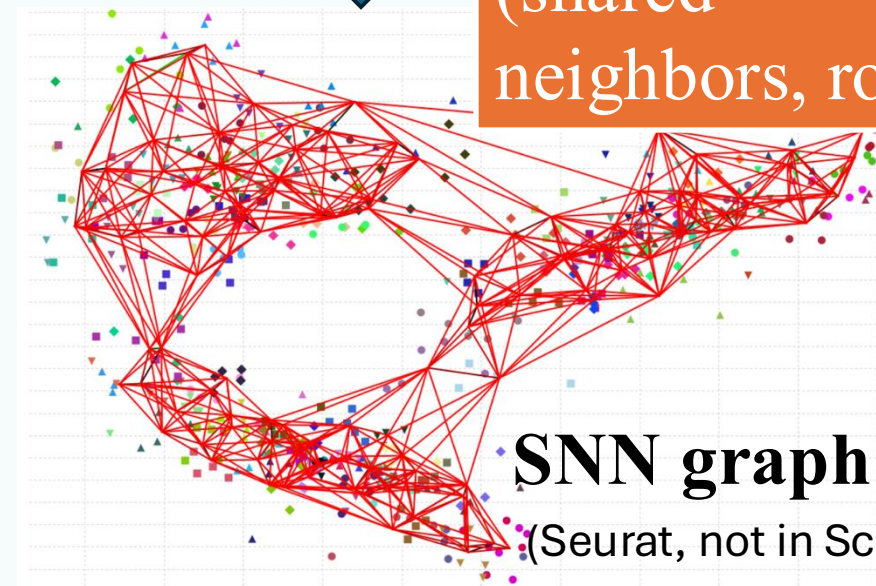
Connect K
nearest
neighbors



KNN graph



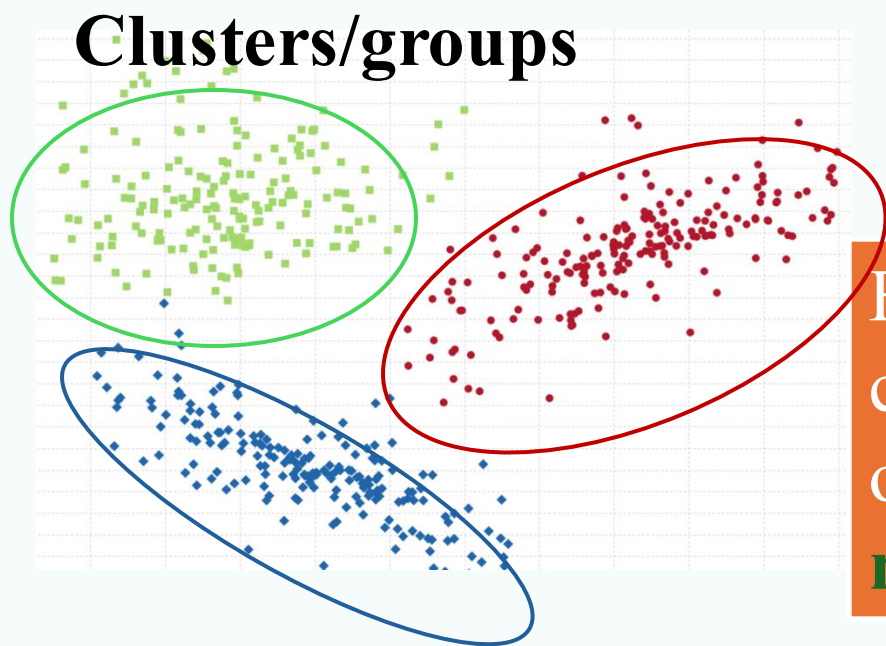
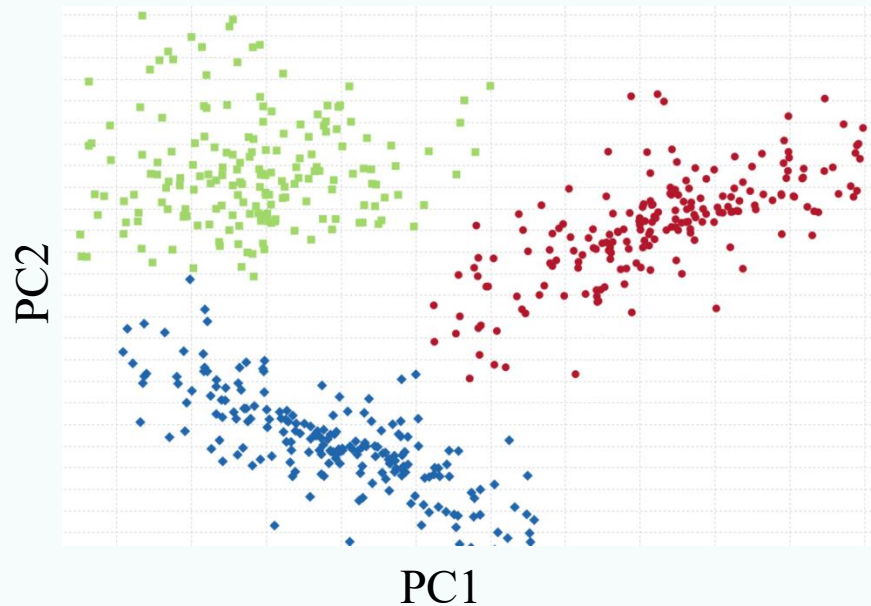
Weight edges
with Jacard index
(shared
neighbors, robust)



SNN graph

(Seurat, not in Scanpy)

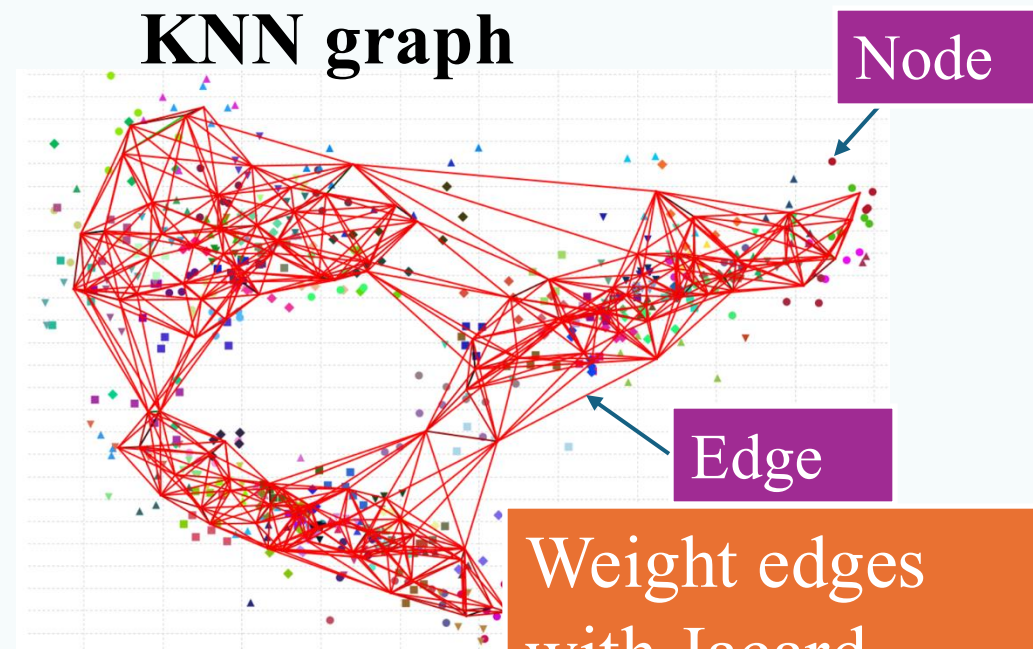
PCA space: on top n PCs



Connect K
nearest
neighbors



KNN graph

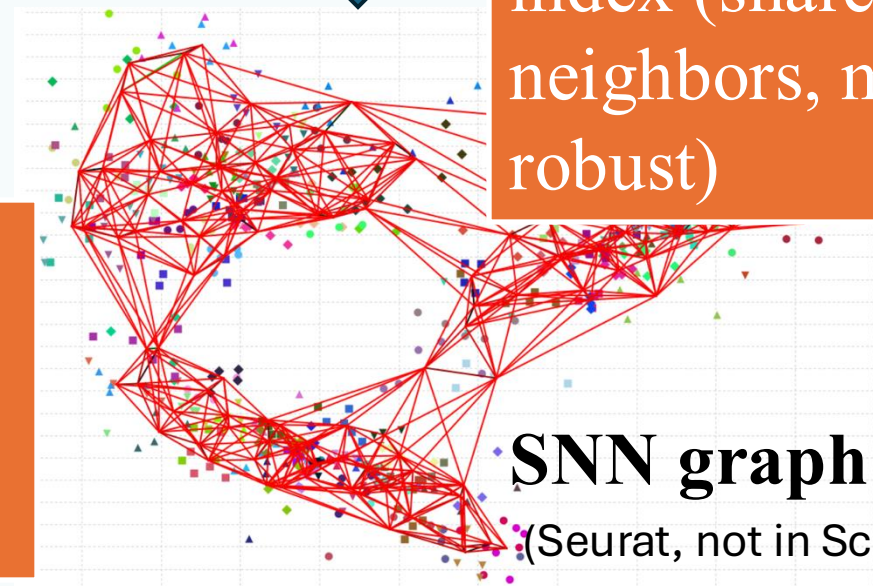


Weight edges
with Jacard
index (shared
neighbors, more
robust)



SNN graph

(Seurat, not in Scanpy)

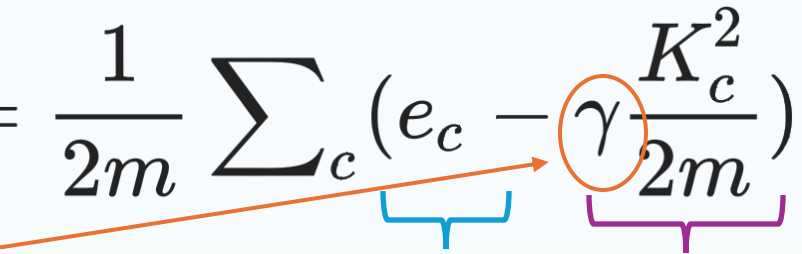


Find cell
communities by
optimizing
modularity



Modularity

Maximize the difference between the observed edges within clusters and the expected edges

$$\mathcal{H} = \frac{1}{2m} \sum_c (e_c - \gamma \frac{K_c^2}{2m})$$


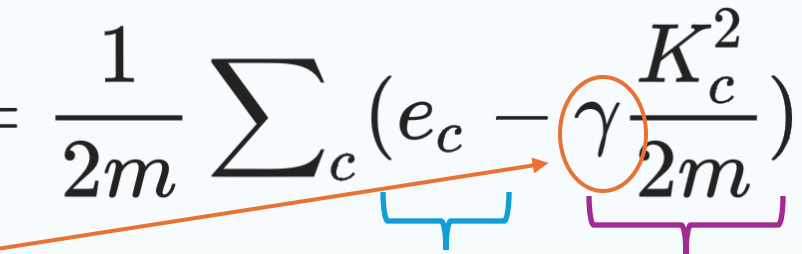
Resolution:
higher, more
clusters

Observed
edges within
cluster c

Expected
edges within
cluster c

Modularity

Maximize the difference between the observed edges within clusters and the expected edges

$$\mathcal{H} = \frac{1}{2m} \sum_c (e_c - \gamma \frac{K_c^2}{2m})$$


Resolution:
higher, more
clusters

Observed
edges within
cluster c

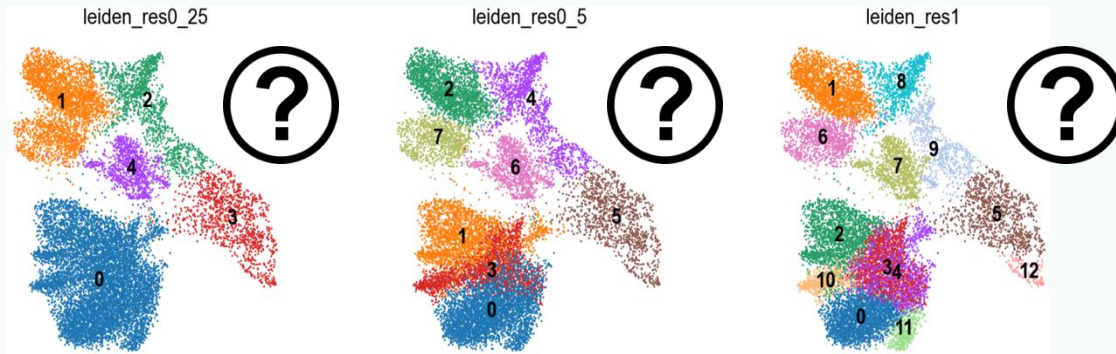
Expected
edges within
cluster c

Modularity optimization algorithms:

Louvain
Leiden

Leiden: an improved version of Louvain

Clustering resolution, quality evaluation



- **Resolution too low:** rare cell types merge with larger clusters, subtypes merge together
- **Resolution too high:** one celltype split to multiple clusters

❖ Measures

- **Silhouette scores**
- Calinski-Harabasz index
- Normalized mutual information
- Variation of information
- Graph-sensitive indices

- ❖ **Cell-based automatic annotation before clustering**
- ❖ The presence of cluster-specific differentially expressed genes
- ❖ Biology

Single cell RNAseq data analysis workflow

Preprocessing
(cellRanger)

Filtering

Normalization

Feature
selection

Scaling

PCA

Visualizaiton:
UMAP/tSNE

Gene set
analysis

Differential analysis:
Clusters/Conditions

Annotation

Clustering

Cluster based annotation

Cell-cell
communication
Cellchat, Nichenet

TF
activity
SCENIC

Trajectory
PAGA
Slingshot

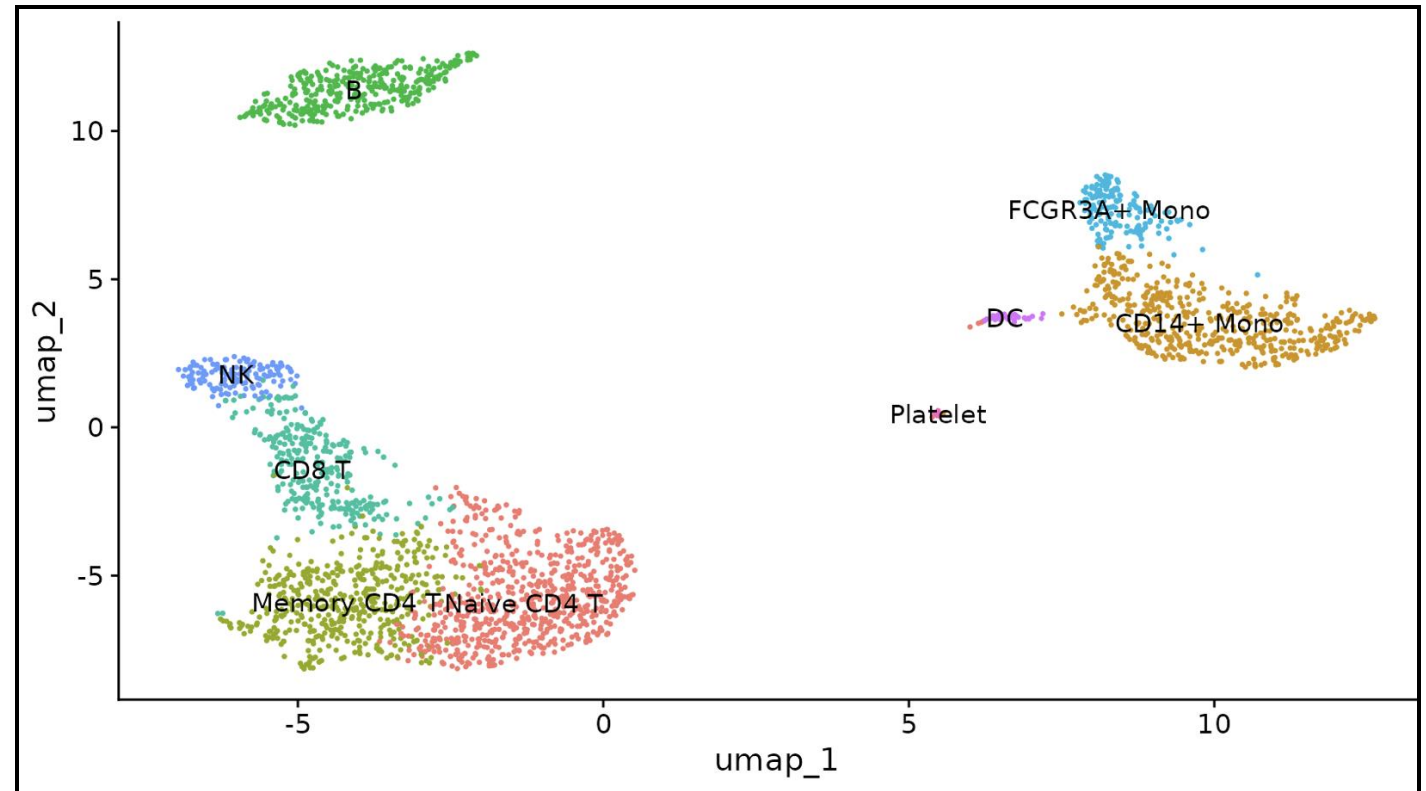
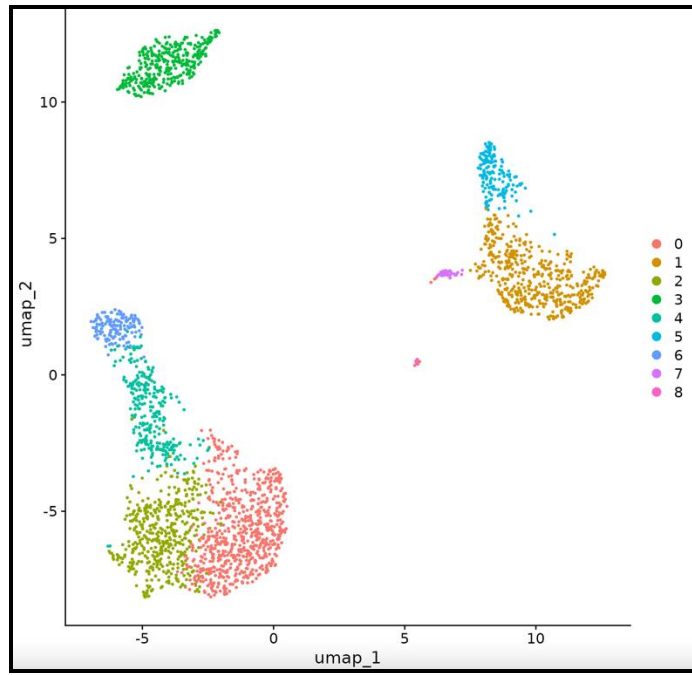
RNA
velocity
scvelo

Metabolic
activity
scCellFie

Cell based annotation

Cell annotation

Cell annotation: the process of labeling cells in your data with celltypes



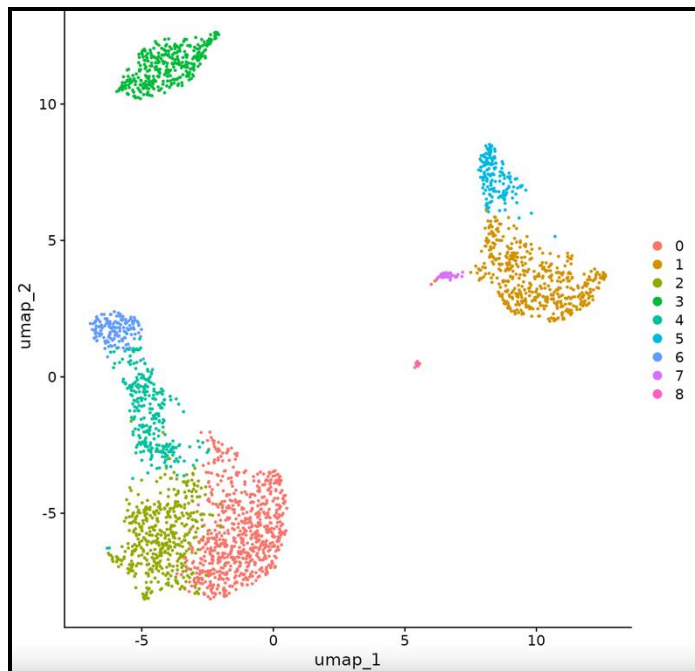
Celltype

- A cellular phenotype that is robust across datasets
- Identifiable based on expression of specific markers (i.e. proteins or gene transcripts)
- Often linked to specific functions
- E.g. a plasma B cell: a type of white blood cell that secretes antibodies used to fight pathogens and it can be identified using specific markers

Manual annotation

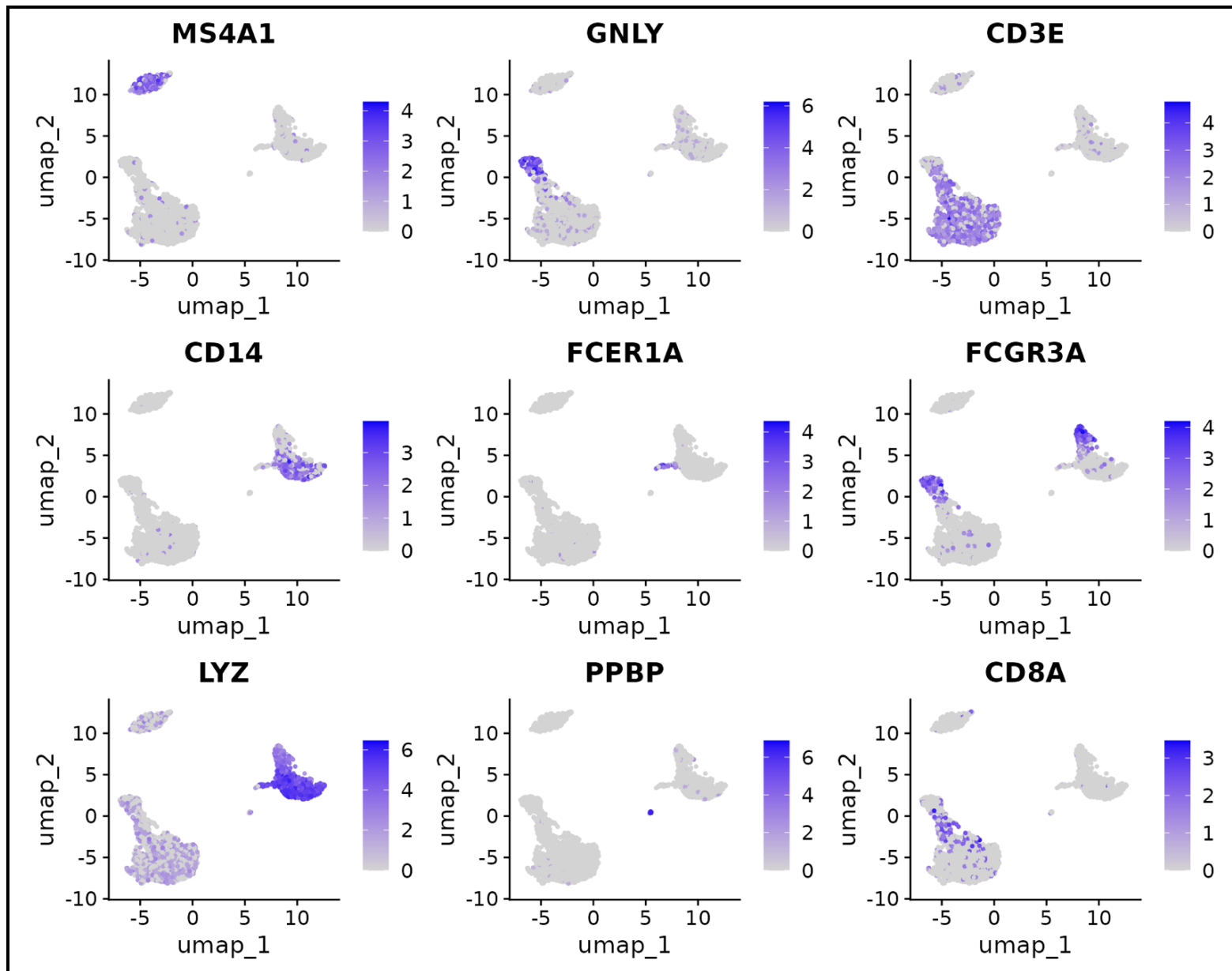
- ❖ Manually annotate cells based on a single or small set of marker genes (*better from RNAseq paper, not from protein study*) known to be associated with a particular cell type.
- ❖ Transparent, novel celltypes
- ❖ **Two angles:**
 - 1. Start from a table of marker genes for all expected celltypes, check their expression in your dataset
 - 2. Check the highly expressed genes/specifically expressed genes in your clusters, see which ones are associated with known celltypes

Manual annotation

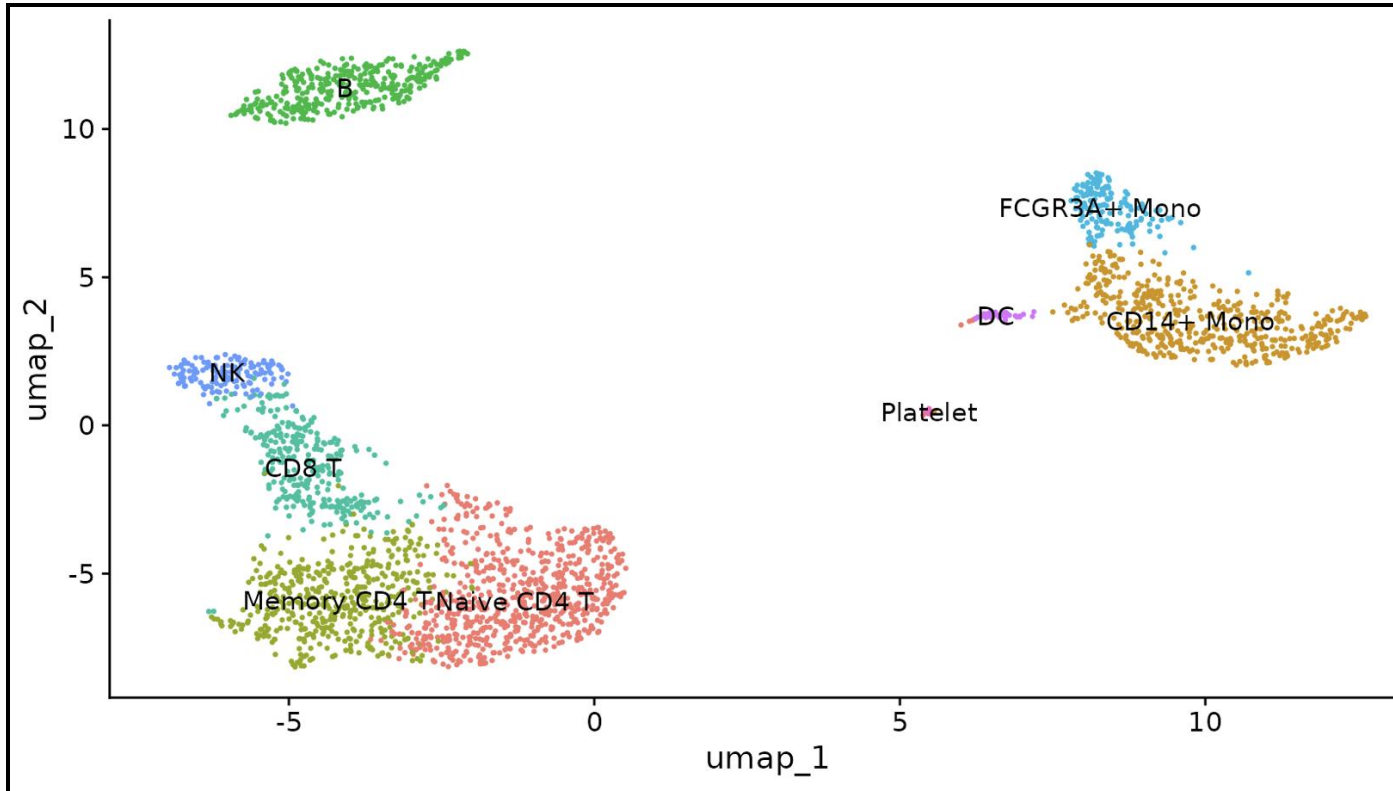


➤ 1. Cluster cells into groups

➤ 2. Check gene expr.



Manual annotation



➤ 3. Assign celltype labels to clusters

Manual annotation

Potential Caveats	Recommendation
Slow, labour-intensive.	Whenever possible, begin with automatic annotation to determine general cell labels.
Subjective.	Work with an expert; consider multiple cell-type conclusions.
Cell types not distinguishable by a single marker.	Use multiple markers for each cell type.
Known markers not distinguishing cell types.	Curate larger lists of markers from the literature, additional experiments or experts.
Conflicting marker gene sets between sources.	Select a marker gene set that best represents the biological signal being looked for in the data (e.g. if looking for cell subtypes, use more extensive gene sets than what is used for general cell-type annotation).



Automatic annotation

- Fast
- Reproducible

Automatic annotation: computer algorithm + prior biological knowledge

- Quickly identify known celltypes
- Highlight unknown celltypes for further investigation (new, subtypes)



Cell annotation

- Ideal (will not miss diff. among cells)
- Low seq. depth: zeros (expressed, not detected), sparse, not enough information single cells

Cell cluster (group) annotation

- More accurate, robust
- Clustering quality
- Compare multiple methods

Automatic annotation: computer algorithm + prior biological knowledge

- Quickly identify known celltypes
- Highlight unknown celltypes for further investigation (new, subtypes)



Marker based

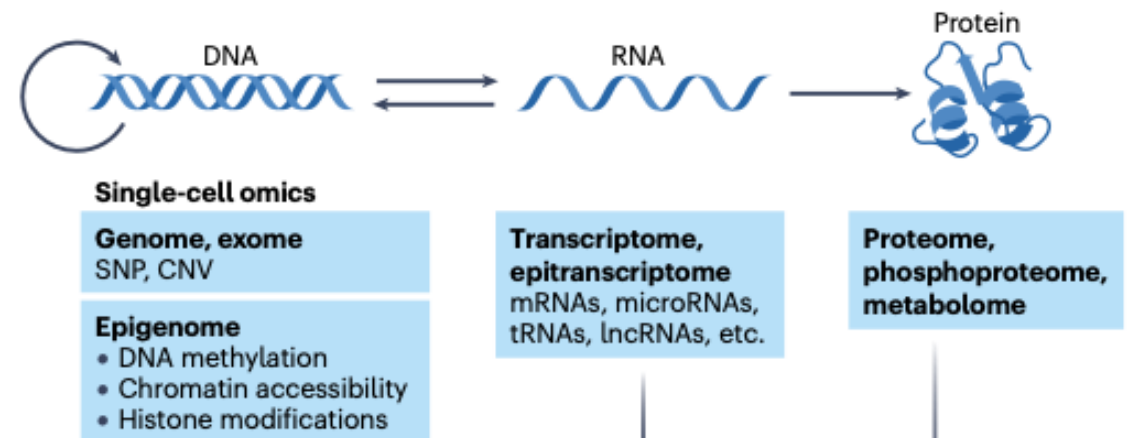
- **Known marker genes sources:** SCSig, PanglaoDB, CellMarker, DISCO, literature
- Conflict between sources
- Tools: AUCell, SCINA, GSEA/GSVA

Reference dataset based

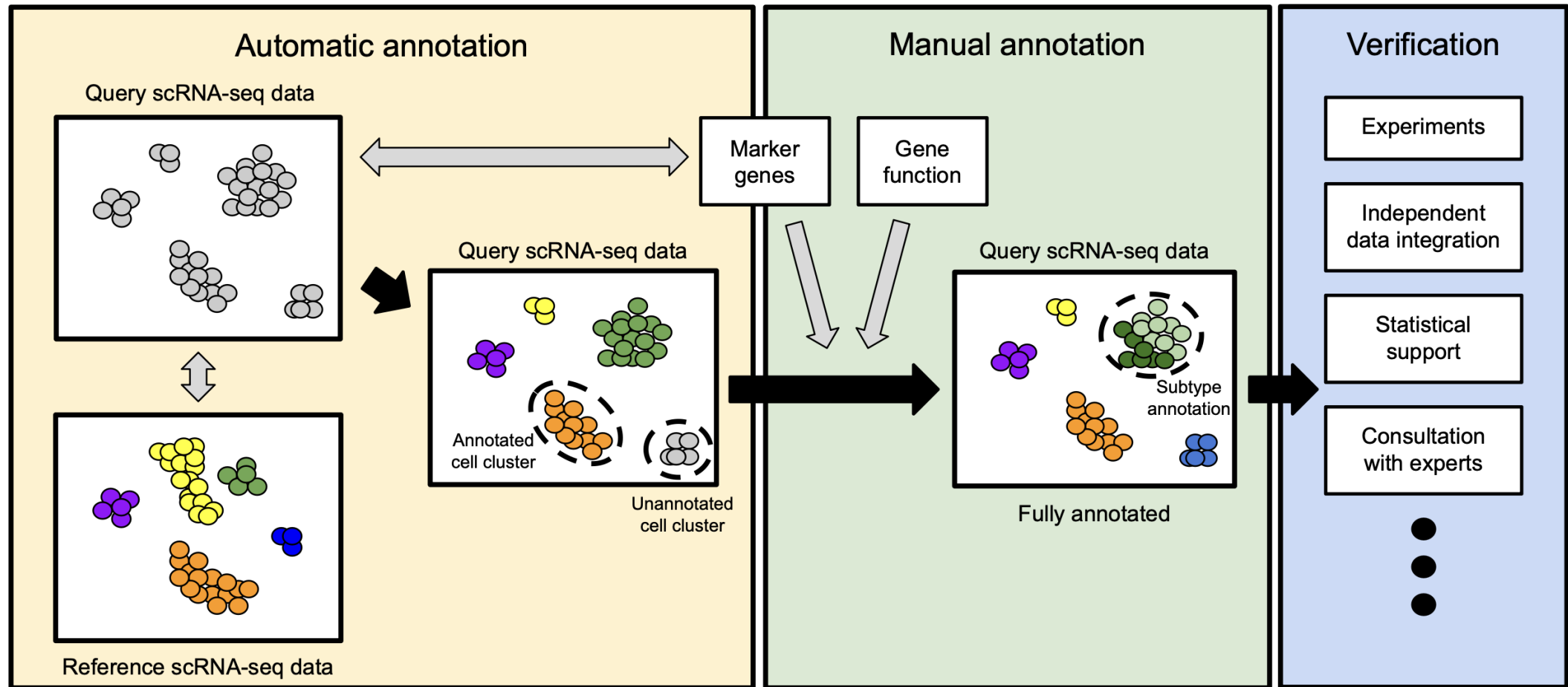
- **Query data:** your data, to be annotated
- **Reference data:** similar, expert-annotated dataset
- **Reference sources:** GEO, Single Cell Expression Atlas, cell atlas projects, DISCO, CellTypist, Azimuth, literature, Gut (<https://www.gutcellatlas.org/pangi.html>), cell ontology (<https://cell-ontology.github.io/>)
- Tools: Scmap, singleCell Net, singleR, Seurat

Annotation verification

- Statistical methods (e.g. Posterior inclusion probabilities, Chung 2020)
- Experimental validation:
 - Functional assays (e.g. cytokine secretion, proliferative capacity),
 - Fluorescence in situ hybridization of source tissue samples,
 - Validate marker gene expression in a larger number of samples
- Complementary genomic methods
 - Proteome, metabolome
 - Genome (SNP)
 - Epigenome
 - Spatial



Recommended workflow for annotation



Differential expression analysis

Single cell RNAseq data analysis workflow

Preprocessing
(cellRanger)

Filtering

Normalization

Feature
selection

Scaling

PCA

Visualizaiton:
UMAP/tSNE

Gene set
analysis

Differential analysis:
Clusters/Conditions

Annotation

Multi-batch
Integration

Clustering

Cluster based annotation

Cell based annotation

Cell-cell
communication
Cellchat, Nichenet

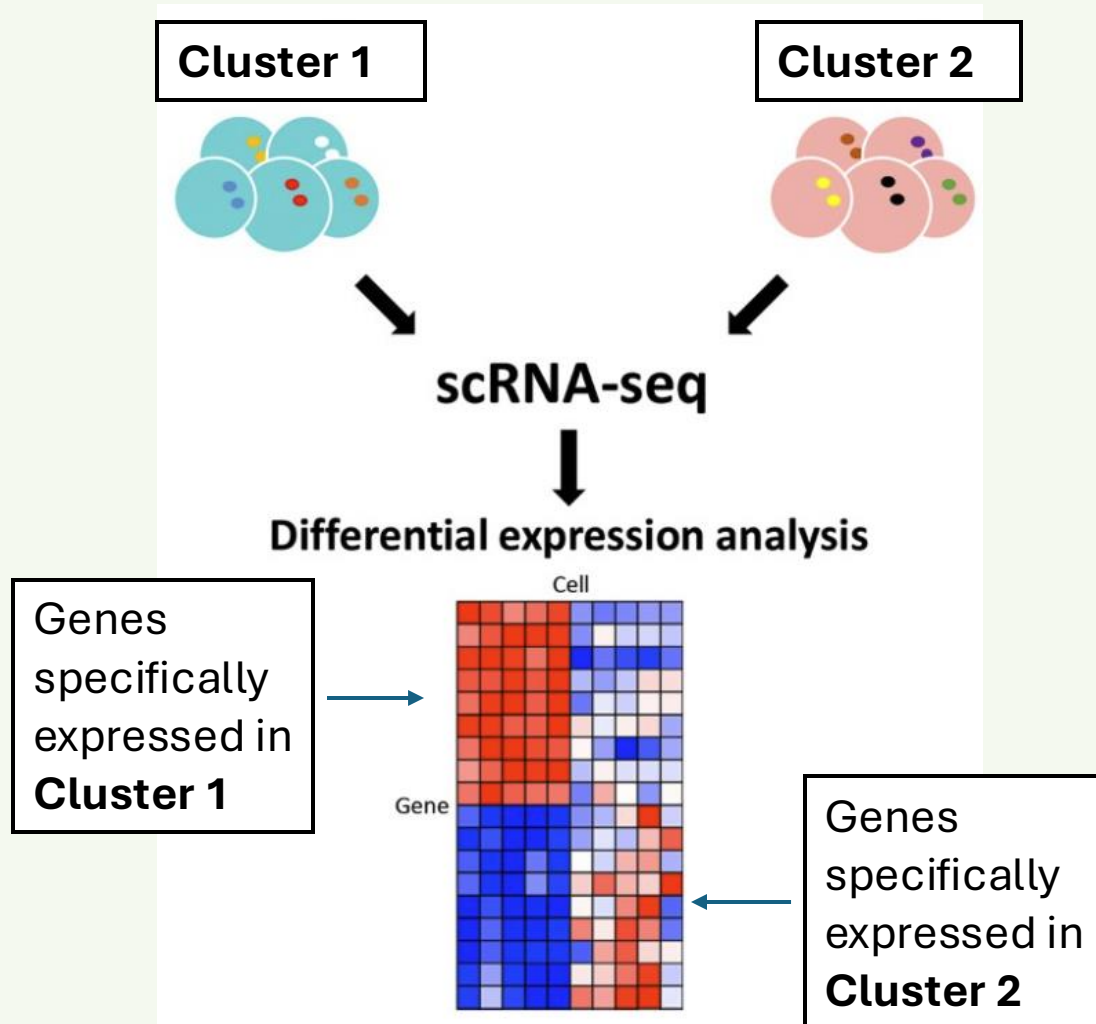
TF
activity
SCENIC

Trajectory
PAGA
Slingshot

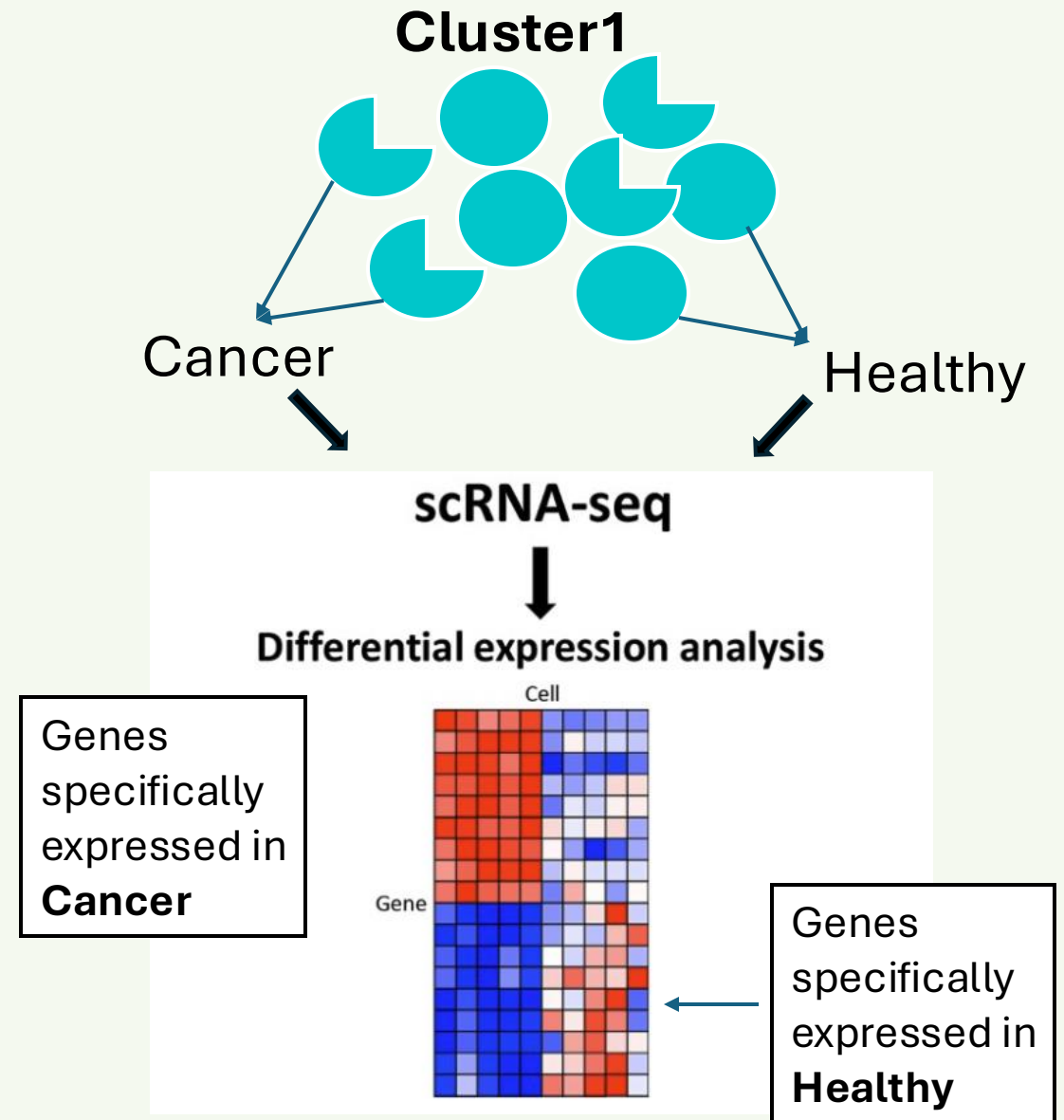
RNA
velocity
scvelo

Metabolic
activity
scCellFie

Between clusters



Between conditions



Methods particularly developed for scRNAseq data:

Single cell RNAseq data:

- Count data (poisson, NB, Guassian...)
- Noisy
- **Zero-inflated** (little material, low seq. depth, sparse, **zero**)
- Bi-modal
- Multi-modal

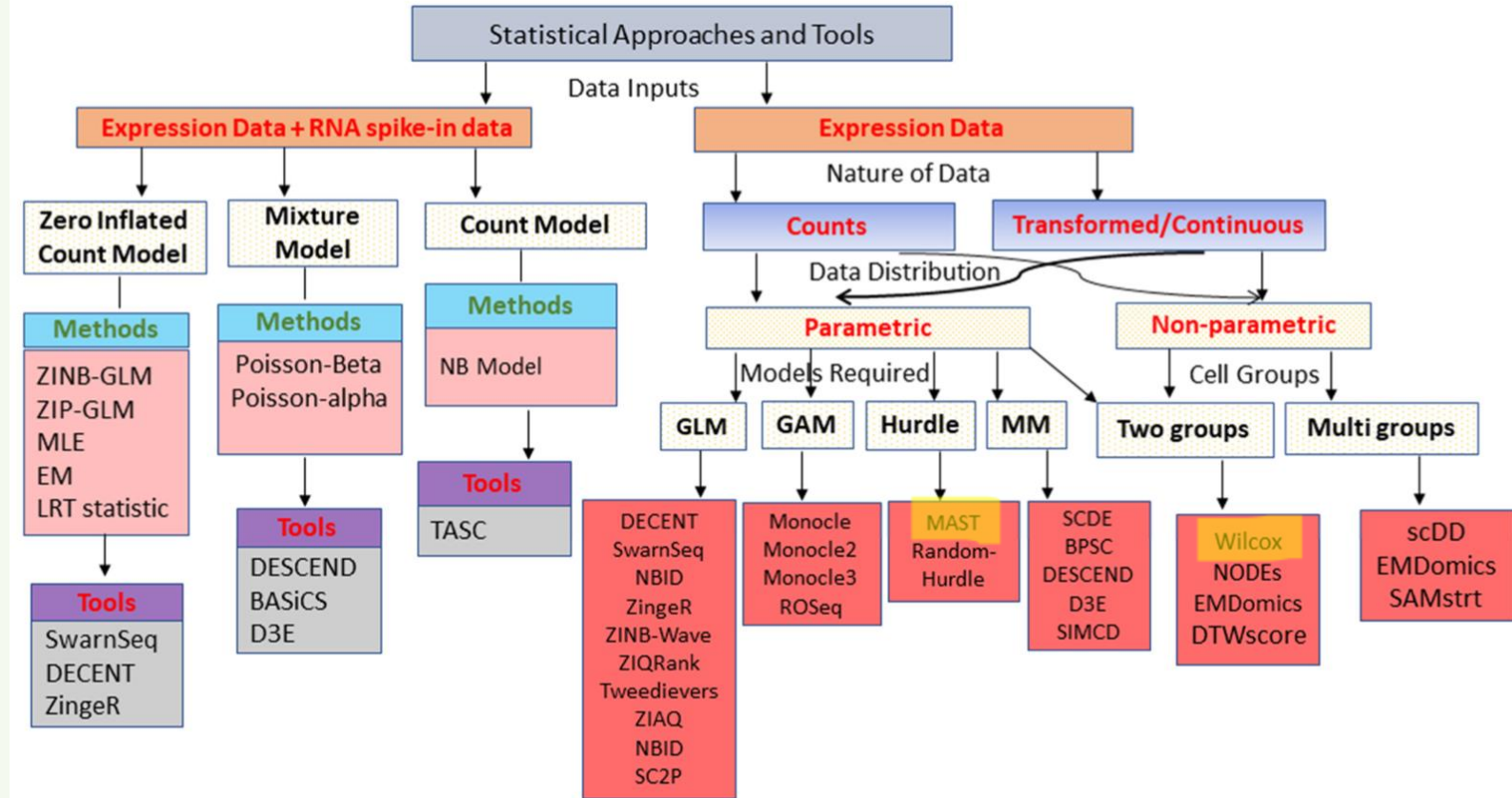
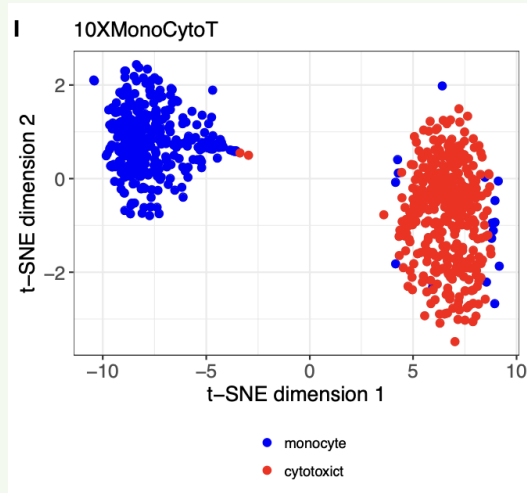
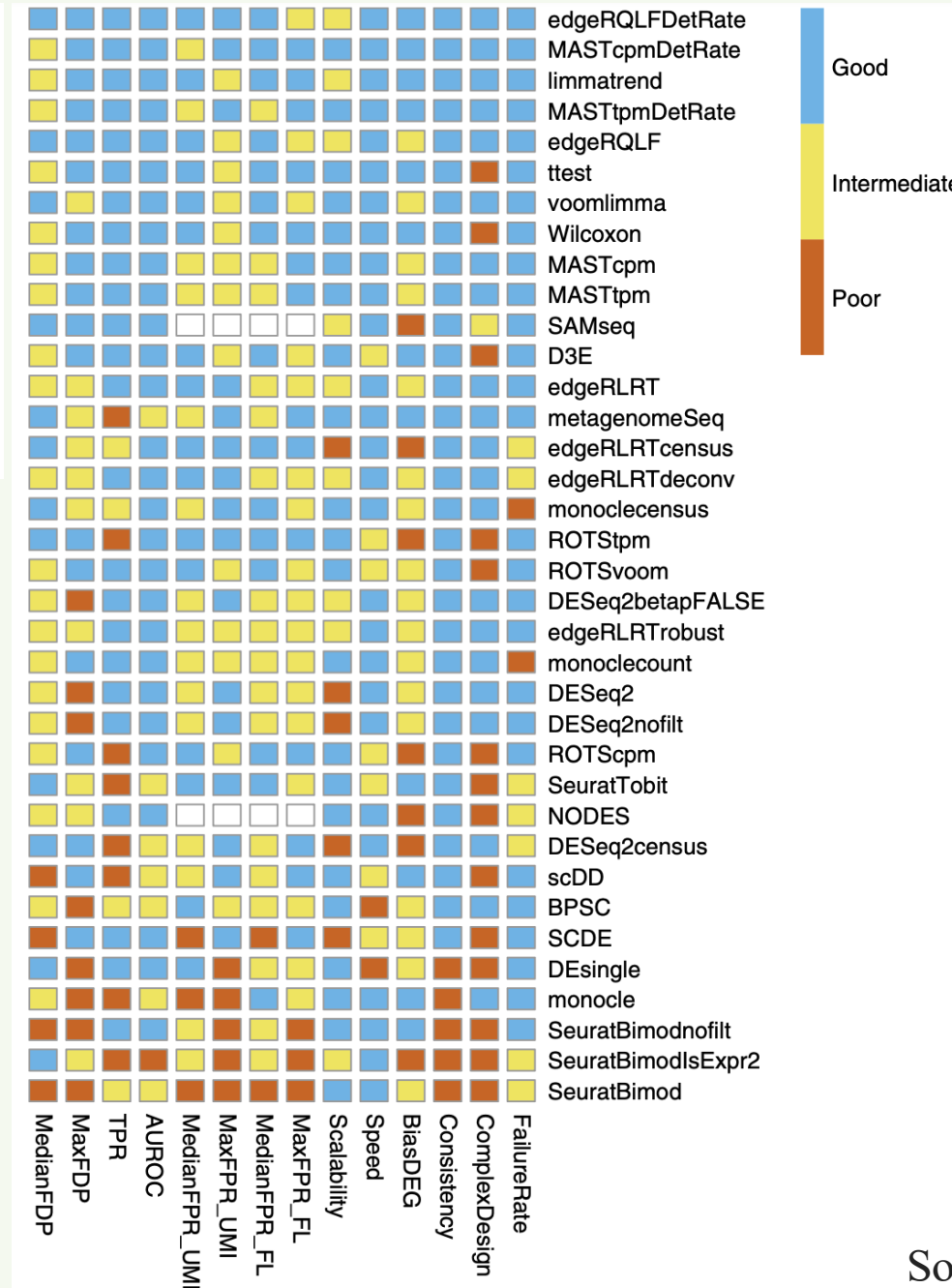


Figure 2. Classification of available statistical approaches and tools used for DEA in single-cell studies. Classification of the approaches is conducted based on the requirement of input data, data

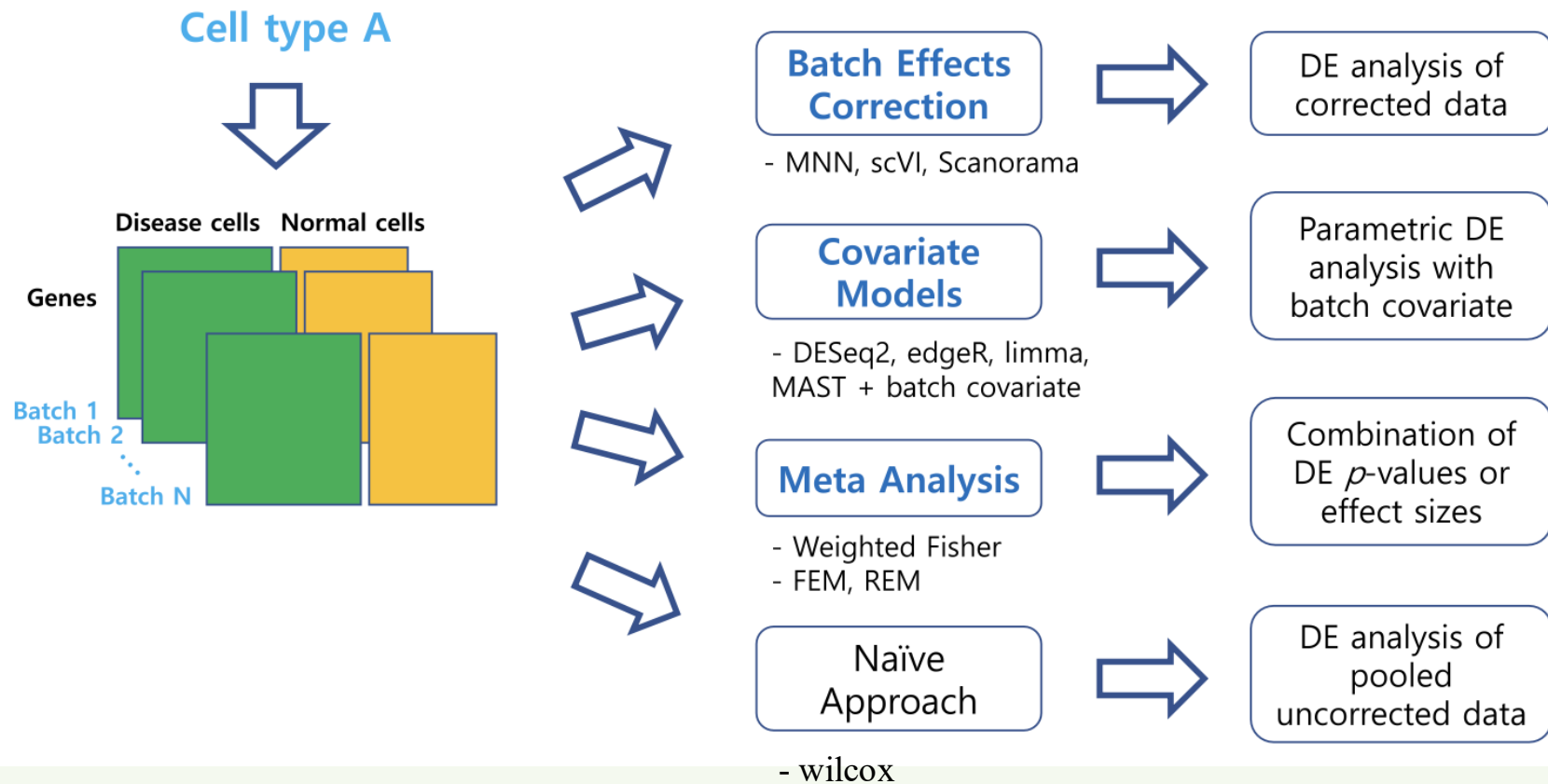
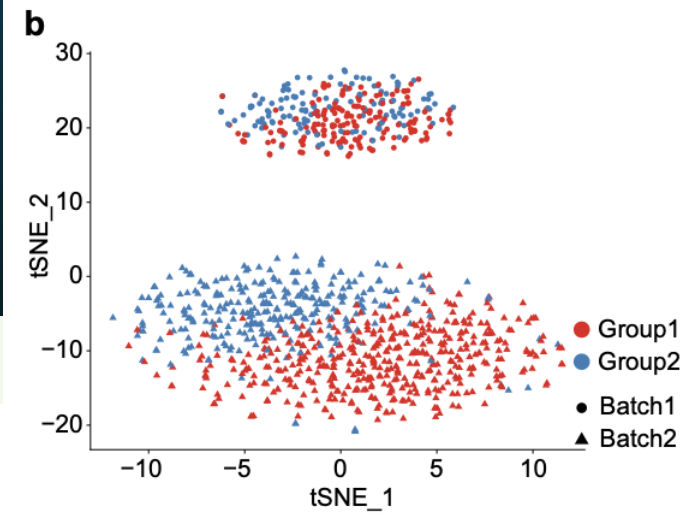


**For simple
comparison
between 2
groups of
cells:
No batches...**

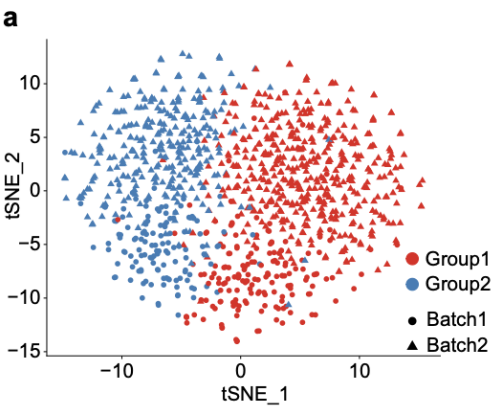


- ❖ t-test and Wilcoxon work well
- ❖ bulk RNA-seq analysis methods (DESeq2, edgeR, Limma...) do not generally perform worse than those specially developed for scRNA-seq

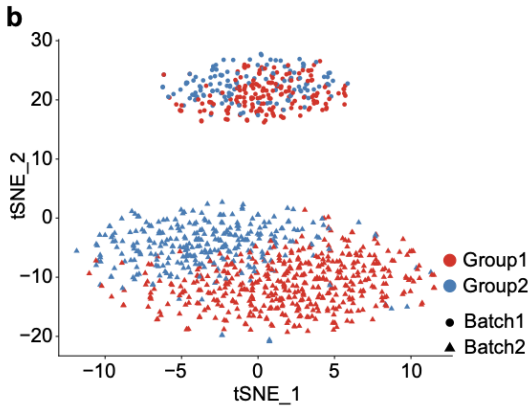
When batches added in



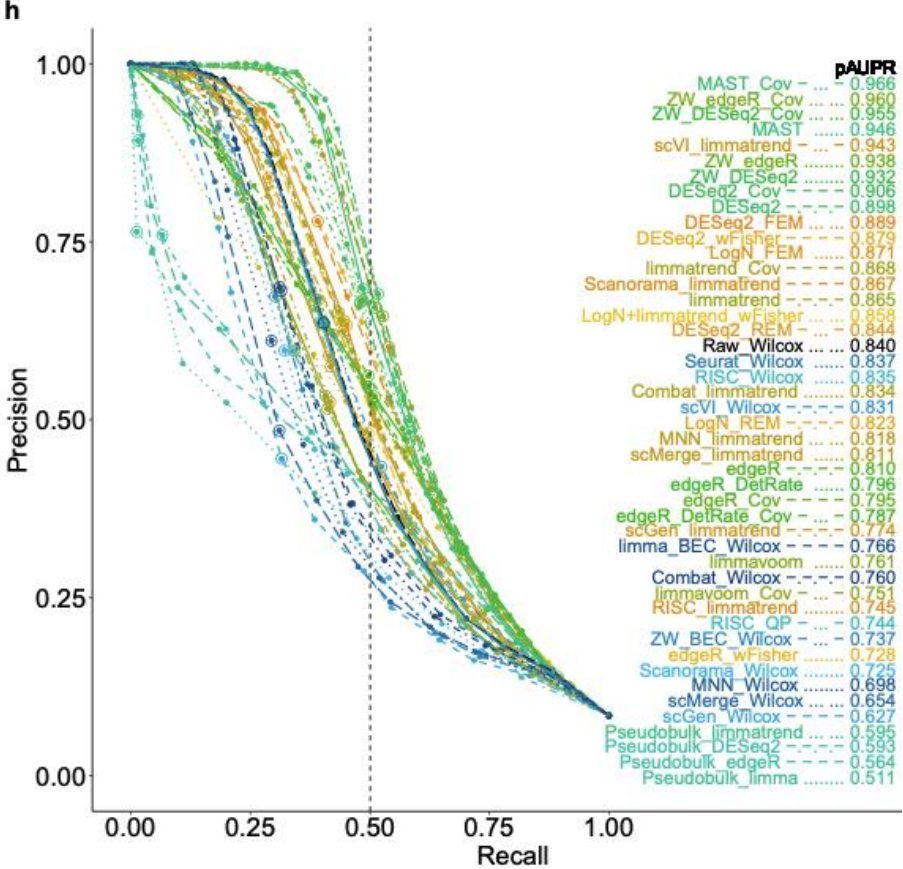
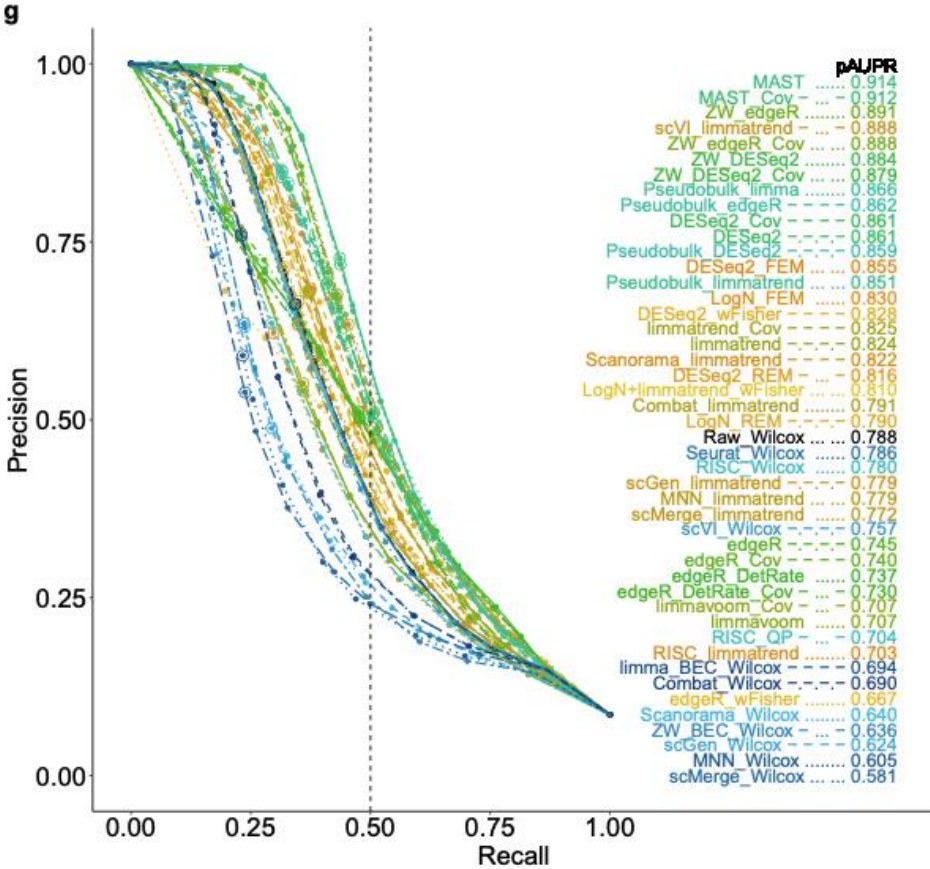
- ❖ Intermediate depth (mean non-zero count of 77)
- ❖ High overall zero rate (> 80%)



Small batch effect



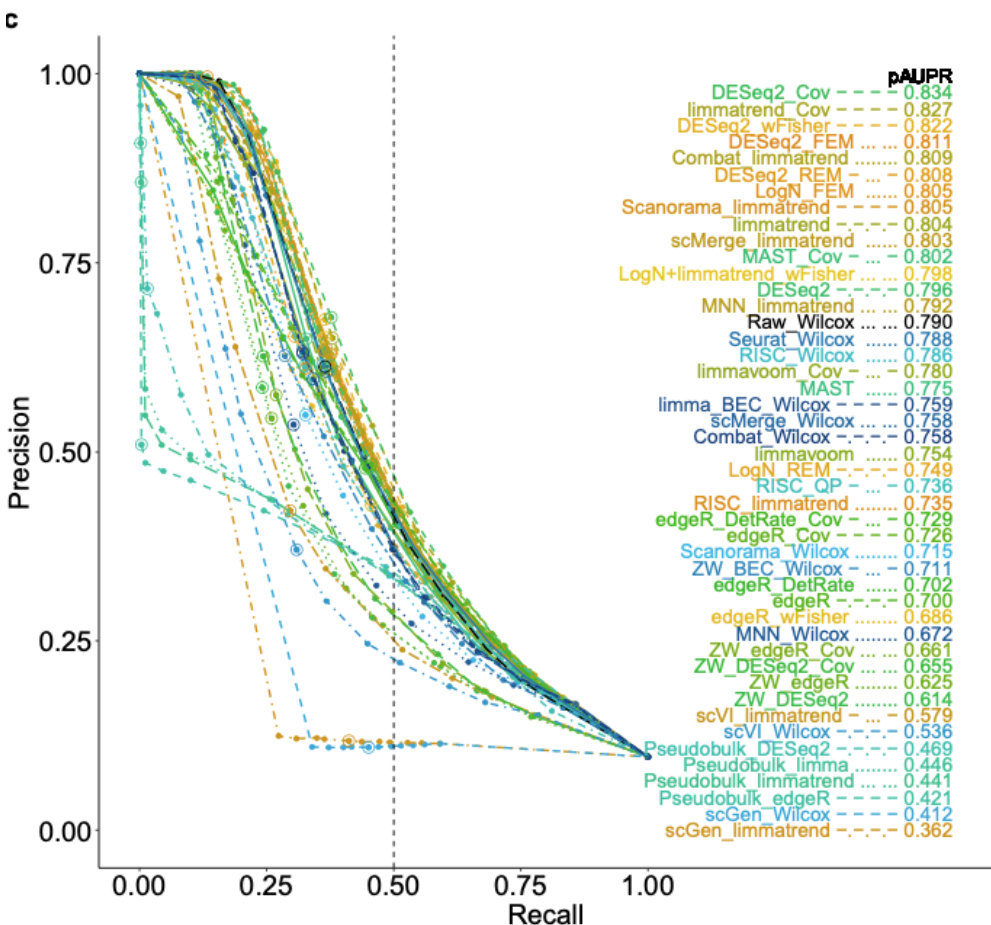
Large batch effect



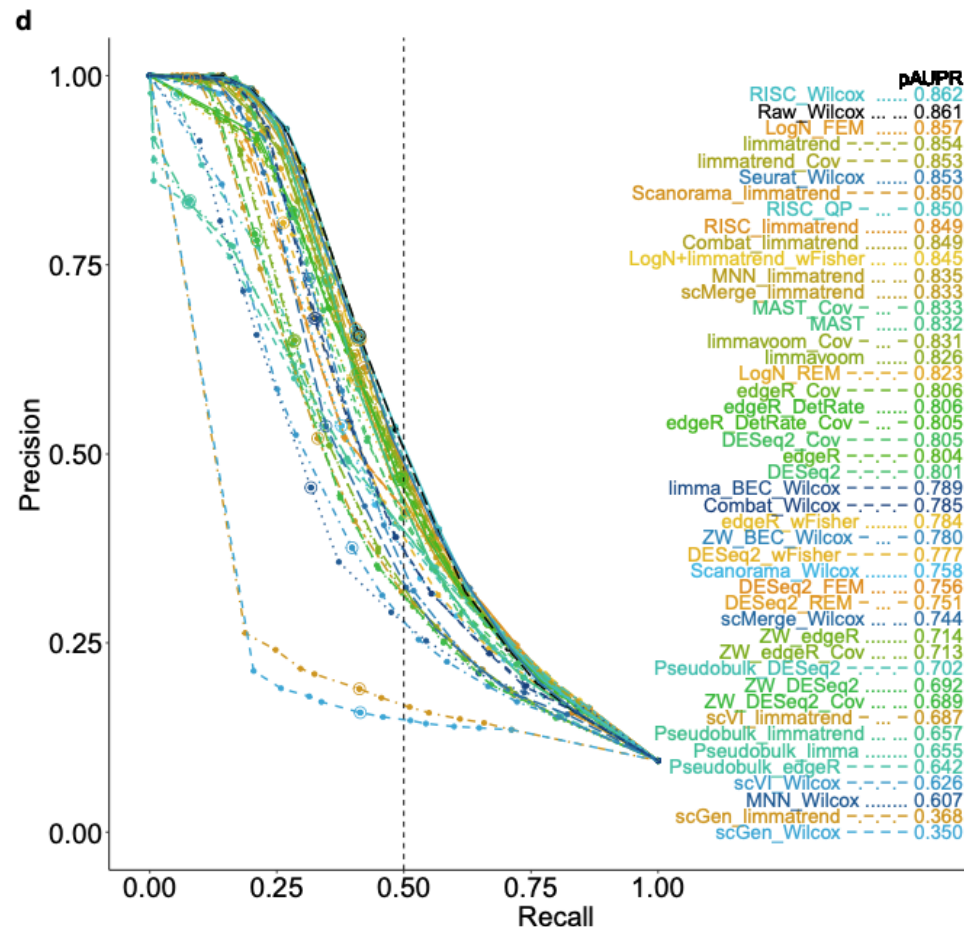
General performance
pAUPR: precision-recall
curve
{precision=TP/(TP+FP),
Recall=TP/(TP+FN)}

- ❖ Low depth (mean non-zero count of 10 & 4/10X Genomics)
- ❖ Zero rate > 80%

Mean non-zero count of 10

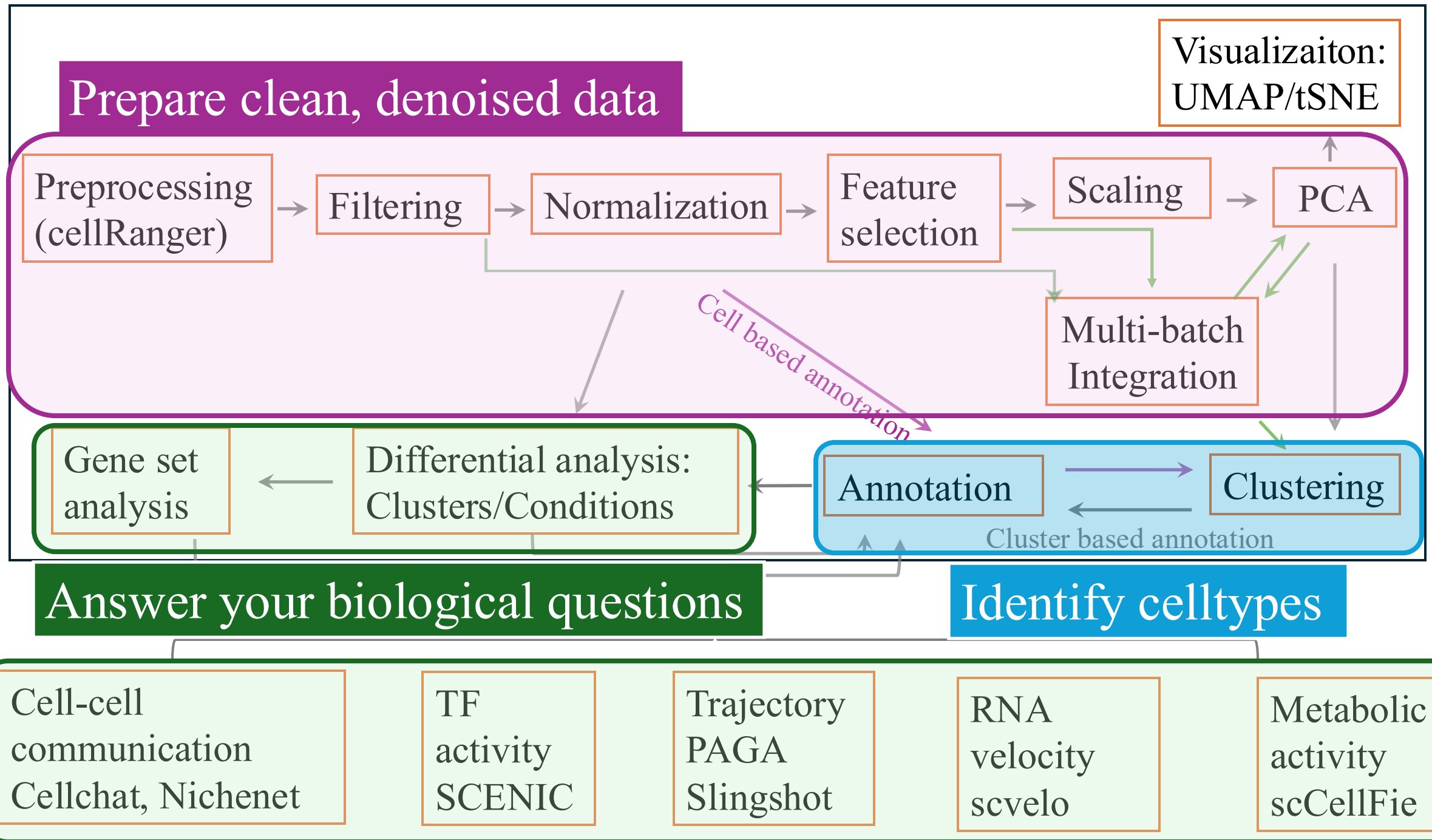


Mean non-zero count of 4 (10X)



General performance
pAUPR: precision-recall
curve
{precision=TP/(TP+FP),
Recall=TP/(TP+FN)}

Basic single cell data analysis summary



References

- **Review:** Andrews, Tallulah S., et al. "Tutorial: guidelines for the computational analysis of single-cell RNA sequencing data." *Nature protocols* 16.1 (2021): 1-9.
- **SCTranform:** Hafemeister, Christoph, and Rahul Satija. "Normalization and variance stabilization of single-cell RNA-seq data using regularized negative binomial regression." *Genome biology* 20.1 (2019): 296.
- **Integration:** Ryu, Yeonjae, et al. "Integration of single-cell RNA-seq datasets: a review of computational methods." *Molecules and cells* 46.2 (2023): 106-119.
- **Integration:** Luecken, Malte D., et al. "Benchmarking atlas-level data integration in single-cell genomics." *Nature methods* 19.1 (2022): 41-50.
- **Dimensional reduction:** Moon, Kevin R., et al. "Visualizing structure and transitions in high-dimensional biological data." *Nature biotechnology* 37.12 (2019): 1482-1492.
- **Clustering:** Traag, Vincent A., Ludo Waltman, and Nees Jan Van Eck. "From Louvain to Leiden: guaranteeing well-connected communities." *Scientific reports* 9.1 (2019): 1-12.
- **Clustering evaluation:** <https://support.sas.com/resources/papers/proceedings19/3409-2019.pdf>
- **Differential analysis:** Soneson, Charlotte, and Mark D. Robinson. "Bias, robustness and scalability in single-cell differential expression analysis." *Nature methods* 15.4 (2018): 255-261.
- **Differential analysis:** Nguyen, Hai CT, et al. "Benchmarking integration of single-cell differential expression." *Nature Communications* 14.1 (2023): 1570.
- **Cell annotation:** Clarke, Zoe A., et al. "Tutorial: guidelines for annotating single-cell transcriptomic maps using automated and manual methods." *Nature protocols* 16.6 (2021): 2749-2764.

